

Radiating-Covariance Controllability of Feathered Wingtip Aeroacoustics

A Theory-Only Mechanism Framework for Segmented Lifting-Surface Directivity

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Abstract

Feathered wingtips and other segmented lifting-surface terminations are often interpreted as acoustic phased arrays, but directivity changes can also arise from source-strength redistribution, local dipole rotation, decoherence, and radiation filtering. This paper develops a theory-only radiating-covariance framework for distinguishing these mechanisms. The far-field spectrum is written as a positive-semidefinite quadratic form, $S_{pp} = \mathbf{a}^H \mathbf{C}_q \mathbf{a}$, and then decomposed into independently radiating source-covariance eigenmodes, $S_{pp} = \sum_r \lambda_r |\mathbf{a}^H \mathbf{u}_r|^2 / (4\pi R)^2$. This identity shows that geometry alone does not determine directivity control; control depends on how source-covariance modes project onto the acoustic radiation operator. Exact computational validation over 1000 randomized covariance cases gives no PSD, Hermitian, modal-identity, or off-diagonal-bound failures. Canonical benchmarks recover diagonal no-phase behavior to 5.8×10^{-16} relative error and a rank-one coherent array factor to 5.6×10^{-16} normalized error. A 625-case partial-coherence sweep produces a mechanism phase diagram in which phase-dominant behavior appears only when geometric phase separation and radiating off-diagonal covariance are both present. Reanalysis of the 50-target feathered-wingtip campaign shows that optimized full-model geometries retain substantial fixed-geometry phase sensitivity ($\overline{\mathcal{P}_R} = 0.811$, phase-ablation RMSE 7.93 dB), while zero-coherence reoptimization remains statistically equivalent to the full model. Thus phase interference is active but not uniquely controlling in the tested regime: the design space can compensate through alternative source-covariance and radiation modes. The released FWA-Bench-0 package provides exact, canonical, transition, and feathered-wingtip benchmark cases for reproducing the computational validation.

1 Introduction

Structured lifting-surface terminations like feathered wingtips are often modeled as acoustic arrays where spatial spacing, sweep, out-of-plane displacement, and incidence vary across elements. While these features suggest bio-inspired noise control through phase interference, directivity changes can also arise from incoherent source-strength redistribution, dipole-axis rotation, decoherence, or radiation filtering. Directivity changes alone therefore do not identify the active mechanism.

This paper addresses this ambiguity by formulating segmented-wingtip directivity control as a radiating-covariance problem. A segmented aeroacoustic surface can control far-field directivity only through the source-covariance modes that project onto the acoustic radiation operator. Phase steering is available only when radiating off-diagonal covariance survives; otherwise, geometry controls directivity through source weighting, local dipole rotation, and incoherent radiation weighting.

The primary contributions of this work are:

1. A positive-semidefinite radiating-covariance identity that decomposes far-field pressure spectra into independently radiating source-covariance eigenmodes.

2. Formal limiting-case theorems for PSD admissibility, diagonal no-phase behavior, rank-one coherent-array behavior, off-diagonal interference bounds, compact phase weakness, and decoherence limits.
3. Radiation-weighted mechanism indices, including raw coherence availability $\mathcal{P}_0$, radiating phase-steerability $\mathcal{P}_R$, radiating covariance fraction $\mathcal{R}$, and spherical-harmonic directivity control dimension D_{control} .
4. Exact computational validation and canonical benchmarks showing recovery of incoherent source-mixer and coherent phased-array limits.
5. A partial-coherence mechanism phase diagram and a reanalysis of the existing 50-target feathered-wingtip campaign through the radiating-covariance metrics.
6. FWA-Bench-0, an open analytical benchmark package containing exact array, diagonal, partial-coherence, and feathered-wingtip cases.

We emphasize that this Version 1 paper is theoretical and computational. It does not use CFD or wind-tunnel data, does not claim absolute broadband noise reduction, and does not claim that feathered wingtips are automatically phase-steering arrays. Its purpose is to make the mechanism-selection theory internally consistent, exactly benchmarked, and reproducible under declared linear far-field stochastic-source assumptions.

2 Literature Review

This work builds upon three established research areas: analytical aeroacoustic loading-noise theory, bio-inspired edge-noise control, and acoustic-array optimization. Classical aerodynamic sound theories, including Curle’s loading noise extension of Lighthill’s analogy [4] and Farassat’s formulations [6], describe acoustic radiation from unsteady surface forces. Trailing-edge and self-noise models [1, 5, 2] establish how turbulent boundary-layer scattering radiates broadband noise. Furthermore, prior studies on serrated edges demonstrate that edge geometry alters directivity and that destructive interference can reduce far-field noise at specific frequencies and angles [3].

Rather than proposing a new acoustic analogy, we implement a closure-based loading-noise surrogate to evaluate relative directivity changes. Because feathered geometries consist of spatially separated, differently oriented source regions, the far-field radiation depends on both local source strength and spatial cross-coherence. While ad-hoc cosine-summed cross-term models can yield non-physical negative power spectral densities (PSDs) under inconsistent coherence closures, we employ a Hermitian positive-semidefinite matrix construction. This formulation guarantees non-negative spectra across all frequencies and observer directions, enabling mathematically consistent comparisons between the zero-coherence, no-delay, and full-coherence limits.

Finally, we utilize thin-airfoil lift and drag proxies as relative screening metrics. These proxies estimate the aerodynamic penalties associated with directivity-optimized mutations without asserting absolute aerodynamic performance.

3 Methodology

3.1 Radiating-covariance operator identity

The central object in this paper is the observer-side quadratic form:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \mathbf{a}^H(\mathbf{n}, \omega) \mathbf{C}_q(\omega) \mathbf{a}(\mathbf{n}, \omega), \quad (1)$$

where $\mathbf{C}_q$ is the source cross-spectral covariance matrix and $\mathbf{a}$ is the radiation vector collecting geometry, propagation phase, transfer, quadrature weight, chord scaling, and local dipole projection. If

$$\mathbf{C}_q(\omega) = \mathbf{U}(\omega)\mathbf{\Lambda}(\omega)\mathbf{U}^H(\omega) = \sum_{r=1}^{N_s} \lambda_r(\omega) \mathbf{u}_r(\omega) \mathbf{u}_r^H(\omega), \quad (2)$$

then Eq. 1 becomes:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \sum_{r=1}^{N_s} \lambda_r(\omega) |\mathbf{a}^H(\mathbf{n}, \omega) \mathbf{u}_r(\omega)|^2. \quad (3)$$

This modal radiation identity is the theoretical core of the paper. The eigenvalue λ_r measures the strength of source covariance mode r , the eigenvector $\mathbf{u}_r$ describes its coherent spatial structure, and $|\mathbf{a}^H \mathbf{u}_r|^2$ measures how efficiently that source mode radiates in observer direction $\mathbf{n}$. Directivity control is therefore governed by the interaction between geometry and radiating source-covariance modes, not by geometry alone.

For an observer set $\{\mathbf{n}_o, w_o\}$, define the integrated radiation operator:

$$\mathbf{W}(\omega) = \sum_o w_o \mathbf{a}_o(\omega) \mathbf{a}_o^H(\omega), \quad (4)$$

where $\mathbf{a}_o = \mathbf{a}(\mathbf{n}_o, \omega)$. The raw off-diagonal coherence availability is:

$$\mathcal{P}_0(\omega) = \frac{\|\mathbf{C}_q - \text{diag}(\mathbf{C}_q)\|_F}{\|\mathbf{C}_q\|_F}. \quad (5)$$

The radiation-weighted phase-steerability index is:

$$\mathcal{P}_R(\omega) = \frac{\|\mathbf{W}^{1/2} \mathbf{C}_{\text{off}} \mathbf{W}^{1/2}\|_F}{\|\mathbf{W}^{1/2} \mathbf{C}_q \mathbf{W}^{1/2}\|_F}, \quad \mathbf{C}_{\text{off}} = \mathbf{C}_q - \text{diag}(\mathbf{C}_q). \quad (6)$$

Unlike $\mathcal{P}_0$, $\mathcal{P}_R$ penalizes coherent covariance that exists in source space but projects weakly onto the far-field radiation operator. With $\mathbf{W}$ normalized to unit mean eigenvalue, we also define the radiating covariance fraction:

$$\mathcal{R}(\omega) = \frac{\text{tr}(\mathbf{W} \mathbf{C}_q)}{\text{tr}(\mathbf{C}_q)}. \quad (7)$$

Finally, directivity is projected onto spherical harmonics:

$$L(\mathbf{n}, \omega) = \sum_{\ell=0}^{L_{\max}} \sum_{m=-\ell}^{\ell} b_{\ell m}(\omega) Y_{\ell m}(\mathbf{n}), \quad (8)$$

and the geometry-to-directivity controllability Jacobian is:

$$J_{\ell m, j} = \frac{\partial b_{\ell m}}{\partial \mu_j}. \quad (9)$$

The directivity control dimension is the numerical rank:

$$D_{\text{control}}(\omega) = \# \{ \sigma_i / \sigma_1 > \epsilon \}, \quad (10)$$

where σ_i are singular values of J and ϵ is a declared tolerance. A target projection score

$$Q_T = \frac{\|P_J T\|_2^2}{\|T\|_2^2} \quad (11)$$

measures how much of a target directivity lies in the locally controllable subspace.

3.2 Physical assumptions and geometry representation

We assume linear, subsonic, far-field acoustics in a uniform medium with density ρ_0 , sound speed c_0 , free-stream speed U_∞ , and Mach number

$$M_\infty = \frac{U_\infty}{c_0}. \quad (12)$$

The observer direction is defined by the unit vector:

$$\mathbf{n} = [n_x \quad n_y \quad n_z]^T, \quad \|\mathbf{n}\| = 1. \quad (13)$$

Feathers are modeled as rigid geometries represented by distributed stochastic loading-noise source lines. Structural vibrations, compliance, and porosity are neglected.

For feather i , the coordinate $\eta \in [0, 1]$ parameterizes the source point:

$$\mathbf{r}_i(\eta; \boldsymbol{\mu}) = \begin{bmatrix} x_i(\eta) \\ y_i(\eta) \\ z_i(\eta) \end{bmatrix}, \quad (14)$$

where $\boldsymbol{\mu}$ is the design vector. The arc length element is $ds_i = J_i(\eta) d\eta$, with Jacobian:

$$J_i(\eta) = \left\| \frac{\partial \mathbf{r}_i}{\partial \eta} \right\|. \quad (15)$$

We estimate J_i via finite differences and apply trapezoidal quadrature.

Root vertical translation, tip sweep, and tip vertical curvature across the feathers $s \in [0, 1]$ are parameterized by quadratic profiles:

$$g(s) = g_1 L_1(s) + g_m L_m(s) + g_7 L_7(s), \quad (16)$$

where g_1 , g_m , and g_7 represent control points for the first, middle, and seventh feathers. The seven feather incidence values α_i are represented in an orthonormal cosine basis with $N_b = N = 7$ modes (one per feather, forming a full-rank square system):

$$\alpha_i = \sum_{m=0}^{N_b-1} B_{im} a_m, \quad (17)$$

where a_m are incidence coefficients and B_{im} is the $N \times N_b$ basis matrix.

3.3 Acoustic model hierarchy

We formulate the aeroacoustic model in three levels to isolate physical assumptions of compactness, spatial distribution, and coherence.

3.3.1 Level 1: compact coherent feather array

Each feather is treated as a compact source at $\mathbf{r}_i$. For observer direction $\mathbf{n}$ and frequency ω , the far-field pressure and deterministic power are:

$$p_1(\mathbf{n}, \omega) = \frac{1}{4\pi R} \sum_{i=1}^N A_i(\mathbf{n}, \omega) q_i(\omega) \exp[ik\mathbf{n} \cdot \mathbf{r}_i], \quad k = \frac{\omega}{c_0}, \quad (18)$$

$$|p_1(\mathbf{n}, \omega)|^2 = \frac{1}{(4\pi R)^2} \left| \sum_{i=1}^N A_i q_i \exp[ik\mathbf{n} \cdot \mathbf{r}_i] \right|^2, \quad (19)$$

where $q_i(\omega)$ is the complex source amplitude, and A_i accounts for local loading and transfer.

3.3.2 Level 2: deterministic distributed source lines

To account for spatial distribution, we integrate along each feather's source line:

$$p_2(\mathbf{n}, \omega) = \frac{1}{4\pi R} \sum_{i=1}^N \int_0^1 H_i(\eta, \mathbf{n}, \omega) q_i(\eta, \omega) \exp[ik\mathbf{n} \cdot \mathbf{r}_i(\eta)] J_i(\eta) d\eta, \quad (20)$$

where H_i collects loading direction and propagation transfer.

3.3.3 Level 3: stochastic cross-spectral model

To represent turbulent loading noise, we define the source cross-spectrum:

$$\Phi_{qq}^{ij}(\eta, \eta', \omega) = \mathbb{E} [q_i(\eta, \omega) q_j^*(\eta', \omega)]. \quad (21)$$

The far-field pressure power spectral density (PSD) becomes:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \sum_{i=1}^N \sum_{j=1}^N \int_0^1 \int_0^1 H_i(\eta, \mathbf{n}, \omega) H_j^*(\eta', \mathbf{n}, \omega) \Phi_{qq}^{ij}(\eta, \eta', \omega) \\ \times \exp[ik\mathbf{n} \cdot (\mathbf{r}_i(\eta) - \mathbf{r}_j(\eta'))] J_i(\eta) J_j(\eta') d\eta d\eta'. \quad (22)$$

Discretizing the quadrature yields the quadratic form:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \mathbf{a}^H(\mathbf{n}, \omega) \mathbf{C}_q(\omega) \mathbf{a}(\mathbf{n}, \omega), \quad (23)$$

where $\mathbf{a}$ aggregates transfer factors, loading orientation, phase, and weights. The source cross-spectral matrix is:

$$\mathbf{C}_q(\omega) = \mathbb{E} [\mathbf{q}(\omega) \mathbf{q}^H(\omega)]. \quad (24)$$

By constructing $\mathbf{C}_q$ as Hermitian positive-semidefinite ($\mathbf{C}_q \succeq 0$), we mathematically guarantee non-negative pressure spectra ($S_{pp} \geq 0$) for all observer angles and frequencies.

3.4 Closure models and nondimensional regime

The source cross-spectral matrix is closed using:

$$C_{q,mn}(\omega) = \sqrt{\Phi_m(\omega) \Phi_n(\omega)} \Gamma_{mn}(\omega), \quad (25)$$

where the source autospectrum is:

$$\Phi_m(\omega) = C_\Phi \rho_0^2 U_\infty^5 c_m^2 F_\chi(\chi_m) F_\alpha(\alpha_m), \quad (26)$$

with C_Φ a scalar calibration constant (not to be confused with the matrix $\mathbf{C}_q$), and:

$$\chi_m = \frac{\omega c_m St_0}{U_c}, \quad U_c = \beta U_\infty, \quad (27)$$

where St_0 is a fixed Strouhal scale parameter that maps chord-based frequency to the spectral shape function F_χ (see Table 12 for its numerical value).

$$F_\chi(\chi) = \frac{\chi^2}{(1 + \chi^2)^{7/3}}, \quad (28)$$

$$F_\alpha(\alpha_m) = 1 + a_\alpha (\alpha_m - \alpha_{\text{ref}})^2. \quad (29)$$

The spatial coherence magnitude and incidence-dependent phase delay are modeled as:

$$|\Gamma_{mn}(\omega)| = \exp \left[-\frac{f}{f_{\text{ref}}} \sqrt{\left(\frac{\Delta x_{mn}}{a_x}\right)^2 + \left(\frac{\Delta y_{mn}}{a_y}\right)^2 + \left(\frac{\Delta z_{mn}}{a_z}\right)^2} \right], \quad (30)$$

$$\tau_\alpha(\alpha_m) = b_\alpha(\alpha_m - \alpha_{\text{ref}}), \quad (31)$$

$$\Gamma_{mn}(\omega) = |\Gamma_{mn}(\omega)| \exp[i\omega(\tau_\alpha(\alpha_m) - \tau_\alpha(\alpha_n))]. \quad (32)$$

The baseline closures ($a_x = a_y = 0.9 \text{ m}$, $a_z = 0.3 \text{ m}$, $f_{\text{ref}} = 1500 \text{ Hz}$, $\beta = 0.85$, and $b_\alpha = 5.0 \times 10^{-4} \text{ s rad}^{-1}$) ensure neighboring feathers remain partially coherent at low-to-mid frequencies while maintaining active phase shifts.

We analyze the system using the center-wing chord $c_{\text{ref}} = 0.30 \text{ m}$ to define nondimensional groups:

$$\tilde{\mathbf{r}} = \frac{\mathbf{r}}{c_{\text{ref}}}, \quad \kappa = kc_{\text{ref}} = \frac{2\pi f c_{\text{ref}}}{c_0}, \quad \tilde{a}_j = \frac{a_j}{c_{\text{ref}}}, \quad \Pi_\alpha = \frac{U_\infty b_\alpha}{c_{\text{ref}}}. \quad (33)$$

The coherence decay is governed by the dimensionless separation:

$$\tilde{d}_{mn} = \sqrt{\left(\frac{\Delta x_{mn}}{a_x}\right)^2 + \left(\frac{\Delta y_{mn}}{a_y}\right)^2 + \left(\frac{\Delta z_{mn}}{a_z}\right)^2}, \quad (34)$$

and the delay phase group:

$$\Theta_{mn} = \omega b_\alpha(\alpha_m - \alpha_n). \quad (35)$$

Table 1: Dimensionless regime parameters for the current study. Baseline nearest-neighbor values are computed from the default geometry using the source discretization employed by the genetic search.

Quantity	Definition	Value in current study
Mach number	$M_\infty = U_\infty/c_0$	0.087
Acoustic size parameter	$\kappa = kc_{\text{ref}}$	1.37–43.96
Convective chord Strouhal	$St_c = f c_{\text{ref}}/U_\infty$	2.5–80
Normalized coherence lengths	$(\tilde{a}_x, \tilde{a}_y, \tilde{a}_z)$	(3.0, 3.0, 1.0)
Delay scale	$\Pi_\alpha = U_\infty b_\alpha/c_{\text{ref}}$	0.05
Baseline nearest inter-feather spacing	$s_{\text{nn}}/c_{\text{ref}}$	0.71–2.35 (median 1.32)
Baseline coherence distance	$\tilde{d}_{\text{nn}}$	0.264–0.924 (median 0.476)

The transfer weight for source point m is:

$$a_m(\mathbf{n}, \omega) = K_H(\mathbf{n}, \omega) c_m (\mathbf{l}_m \cdot \mathbf{n}) w_m \exp[ik\mathbf{n} \cdot \mathbf{r}_m], \quad (36)$$

with kernel:

$$K_H(\mathbf{n}, \omega) = \frac{\omega/c_0}{(1 - M_\infty n_x)^2}. \quad (37)$$

3.5 Surrogate validation, target fit, and aerodynamic proxies

We validate optimized geometries using a separate boundary-layer self-noise surrogate simulator. This simulator uses BPM/TNO-inspired spectral closures (trailing-edge, separation, bluntness, and tip-vortex contributions) rather than the optimizer’s compact loading-noise closures. The local aerodynamic state relies on a thin-airfoil lift proxy with soft stall:

$$C_l = 2\pi\alpha \left[1 + \left(\frac{|\alpha|}{\alpha_{\text{stall}}} \right)^8 \right]^{-1}, \quad (38)$$

and drag coefficient $C_d = C_{d0} + k_d C_l^2$.

Target directivity patterns T_o are generated with $N_\ell = 1, 2$, or 3 smooth lobes:

$$T_o = \sum_{\ell=1}^{N_\ell} A_\ell \left[\frac{1 + \mathbf{n}_o \cdot \mathbf{c}_\ell}{2} \right]^{p_\ell}. \quad (39)$$

The optimizer minimizes the mean-squared error of peak-normalized directivities:

$$J_{\text{fit}} = \frac{1}{N_o} \sum_{o=1}^{N_o} \left(\hat{S}_o^{\text{dB}} - \hat{T}_o^{\text{dB}} \right)^2. \quad (40)$$

We report target-fit and direction errors:

$$E_{\text{RMSE}} = \left[\frac{1}{N_o} \sum_{o=1}^{N_o} \left(\hat{S}_o^{\text{dB}} - \hat{T}_o^{\text{dB}} \right)^2 \right]^{1/2}, \quad (41)$$

$$E_{\text{MAE}} = \frac{1}{N_o} \sum_{o=1}^{N_o} \left| \hat{S}_o^{\text{dB}} - \hat{T}_o^{\text{dB}} \right|, \quad (42)$$

$$E_{\text{peak}} = \cos^{-1} (\mathbf{n}_{\text{target peak}} \cdot \mathbf{n}_{\text{predicted peak}}). \quad (43)$$

Aerodynamic performance is tracked using integrated lift and drag proxies:

$$L^* = \sum_m \frac{1}{2} \rho_0 U_\infty^2 C_{l,m} c_m \Delta s_m, \quad (44)$$

$$D^* = \sum_m \frac{1}{2} \rho_0 U_\infty^2 C_{d,m} c_m \Delta s_m, \quad (45)$$

and the ratio $(L/D)^* = L^*/D^*$. Baseline-relative retention ratios R_L , R_D , and $R_{L/D}$ measure the aerodynamic impact of structural mutations.

3.6 Mechanism ablation and optimization campaign

We isolate acoustic mechanisms by optimizing each target under four distinct configurations:

1. **Incidence only** (incidence): Spatial parameters are fixed; only the 7 incidence coefficients are optimized.
2. **Full geometry, zero coherence** (zero_coherence): The full 18-variable vector is optimized, but $\Gamma_{mn} = \delta_{mn}$, disabling coherent phase-steering.
3. **Full geometry, no delay** (no_delay): The full 18-variable vector is optimized, but $\tau_\alpha = 0$, disabling incidence-delay phase shifts.
4. **Full model** (full): The full 18-variable design space is optimized under the complete model.

The genetic optimizer searches the 18 design variables using a differential-evolution population of size:

$$P = N_p N_\mu, \quad (46)$$

where $N_\mu = 18$ and $N_p = 9$ (the population multiplier). Candidates are penalized for physical source-line intersections. We employ a two-stage evaluation: candidates are screened on a coarse grid (observer stride 2, frequency stride 2, $n_\eta = 5$), and the top fraction (18%, with a minimum of 6 candidates) is evaluated at the full-resolution observer and frequency grid. The mutation factor is drawn uniformly from $[0.5, 1.0]$ per generation, the crossover probability is 0.7, and early stopping is triggered after 20 consecutive generations without improvement exceeding 0.015 dB².

To accelerate the genetic search, the PyTorch acoustic path refactors the CSD matrix projection to scale transfer weights $\mathbf{a}_m$ directly by $\sqrt{\Phi_m}$. This changes the memory footprint from $O(N^2)$ to $O(N)$, and maps tensor contractions to hardware matrix units (AMX/MPS). For the `zero_coherence` ablation, a specialized fast path skips distance and matrix calculations entirely, evaluating far-field pressure power as $S_{pp}(\mathbf{n}, \omega) = \sum_m |a_m|^2$.

3.7 Statistical analysis

We perform a paired repeated-target analysis across 50 random target seeds. Pairwise contrasts $d_s^{(A,B)} = y_s^{(A)} - y_s^{(B)}$ for response variables y (validated RMSE, drag ratio, and L/D retention) are evaluated using two-sided paired Student- t tests. Family-wise error rates are controlled using Holm-adjusted p -values.

4 Mathematical and Numerical Validation

To ensure the physical validity, numerical robustness, and interpretability of the radiating-covariance framework, this section combines formal limiting-case propositions with exact computational validation. The propositions establish PSD admissibility, Hermitian symmetry, covariance eigenmode radiation, diagonal no-phase behavior, rank-one coherent-array recovery, off-diagonal interference bounds, compact/decoherence limits, and directivity controllability limits. The subsequent numerical checks verify grid convergence, calibration-shape invariance, and the phase-mechanism interpretation of the optimization campaign.

4.1 Proposition 1 — Power Spectral Density Non-negativity

Let the source cross-spectral matrix $\mathbf{C}_q(\omega) \in \mathbb{C}^{N \times N}$ be positive-semidefinite (PSD), denoted as $\mathbf{C}_q(\omega) \succeq 0$. We show that the far-field pressure power spectral density $S_{pp}(\mathbf{n}, \omega) \geq 0$ for all frequencies $f = \omega/(2\pi) \geq 0$ and observer directions $\mathbf{n}$. This proof relies on the physical assumptions that the observer distance satisfies $R > 0$, the reference frequency satisfies $f_{\text{ref}} > 0$, the coherence decay parameters satisfy $a_x, a_y, a_z > 0$, the source autospectra satisfy $\Phi_m(\omega) \geq 0$, and the frequency parameter satisfies $f \geq 0$.

Proof. Recall the discretized far-field pressure PSD given by Eq. 23:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \mathbf{a}^H(\mathbf{n}, \omega) \mathbf{C}_q(\omega) \mathbf{a}(\mathbf{n}, \omega).$$

By definition of a positive-semidefinite matrix, for any complex vector $\mathbf{z} \in \mathbb{C}^N$, the quadratic form satisfies:

$$\mathbf{z}^H \mathbf{C}_q(\omega) \mathbf{z} \geq 0.$$

Since $\mathbf{a}(\mathbf{n}, \omega) \in \mathbb{C}^N$ is a well-defined complex transfer weight vector, and since the propagation distance factor $R > 0$ yields a positive real denominator $(4\pi R)^2 > 0$, substituting $\mathbf{z} = \mathbf{a}(\mathbf{n}, \omega)$ directly yields:

$$S_{pp}(\mathbf{n}, \omega) \geq 0.$$

This guarantees that the predicted acoustic power remains non-negative under any observer angle and frequency, preventing non-physical negative energy states. ■

We now prove that each of the three implemented coherence models yields $\mathbf{C}_q(\omega) \succeq 0$. The source cross-spectral matrix is closed via Eq. 25 as:

$$C_{q,mn}(\omega) = \sqrt{\Phi_m(\omega)\Phi_n(\omega)}\Gamma_{mn}(\omega).$$

Let $\mathbf{v} = [\sqrt{\Phi_1}, \dots, \sqrt{\Phi_N}]^T \in \mathbb{R}^N$ be a real-valued vector of source amplitude terms (since $\Phi_i \geq 0$). We define the rank-1 outer product matrix $\mathbf{S} = \mathbf{v}\mathbf{v}^T \in \mathbb{R}^{N \times N}$. For any complex vector $\mathbf{z} \in \mathbb{C}^N$:

$$\mathbf{z}^H \mathbf{S} \mathbf{z} = \mathbf{z}^H (\mathbf{v}\mathbf{v}^T) \mathbf{z} = (\mathbf{z}^H \mathbf{v})(\mathbf{v}^T \mathbf{z}) = |\mathbf{z}^H \mathbf{v}|^2 \geq 0.$$

Hence, $\mathbf{S} \succeq 0$. The source cross-spectral matrix can be written as the Hadamard (entrywise) product:

$$\mathbf{C}_q = \mathbf{S} \circ \mathbf{\Gamma},$$

where $\mathbf{\Gamma} \in \mathbb{C}^{N \times N}$ is the spatial coherence matrix containing Γ_{mn} . By the Schur Product Theorem, the Hadamard product of two positive-semidefinite matrices is also positive-semidefinite. Therefore, to show $\mathbf{C}_q \succeq 0$, it is sufficient to prove that the spatial coherence matrix $\mathbf{\Gamma} \succeq 0$ for each implemented model.

1. Zero Coherence Model. Under the `zero_coherence` limit, the coherence elements are:

$$\Gamma_{mn} = \delta_{mn},$$

which corresponds to the identity matrix $\mathbf{\Gamma} = \mathbf{I}$. Because all eigenvalues of the identity matrix are unity ($1 > 0$), $\mathbf{I}$ is positive-definite, and hence $\mathbf{\Gamma} \succeq 0$.

2. Full Coherence Model. In the completely coherent limit, the complex coherence function is:

$$\Gamma_{mn}(\omega) = \exp[i\omega(\tau_\alpha(\alpha_m) - \tau_\alpha(\alpha_n))].$$

Defining the complex vector $\mathbf{u} \in \mathbb{C}^N$ with elements $u_m = \exp[i\omega\tau_\alpha(\alpha_m)]$, the coherence matrix can be expressed as the rank-1 outer product $\mathbf{\Gamma} = \mathbf{u}\mathbf{u}^H$. For any $\mathbf{z} \in \mathbb{C}^N$:

$$\mathbf{z}^H \mathbf{\Gamma} \mathbf{z} = \mathbf{z}^H (\mathbf{u}\mathbf{u}^H) \mathbf{z} = |\mathbf{z}^H \mathbf{u}|^2 \geq 0.$$

Thus, $\mathbf{\Gamma} \succeq 0$.

3. Exponential Coherence Model. For the partially coherent case, the coherence matrix incorporates spatial decay:

$$\Gamma_{mn}(\omega) = \exp\left[-\tilde{d}_{mn} \frac{f}{f_{\text{ref}}}\right] \exp[i\omega(\tau_\alpha(\alpha_m) - \tau_\alpha(\alpha_n))].$$

Let $\mathbf{D} \in \mathbb{R}^{N \times N}$ be the real spatial correlation matrix with elements:

$$D_{mn} = \exp\left[-\tilde{d}_{mn} \frac{f}{f_{\text{ref}}}\right],$$

where $\tilde{d}_{mn}$ is the dimensionless distance metric given by Eq. 34. The matrix $\mathbf{\Gamma}$ can be written as the Hadamard product of $\mathbf{D}$ and the full coherence rank-1 outer product:

$$\mathbf{\Gamma} = \mathbf{D} \circ (\mathbf{u}\mathbf{u}^H).$$

By the Schur Product Theorem, since $\mathbf{u}\mathbf{u}^H \succeq 0$, it suffices to show $\mathbf{D} \succeq 0$. The elements D_{mn} are evaluated as:

$$D_{mn} = \exp \left[-c \|\mathbf{x}'_m - \mathbf{x}'_n\|_2 \right],$$

where $c = f/f_{\text{ref}} \geq 0$ is a non-negative scalar, and the scaled coordinate vector of source point m is $\mathbf{x}'_m = [x_m/a_x, y_m/a_y, z_m/a_z]^T \in \mathbb{R}^3$ (which is well-defined under $a_x, a_y, a_z > 0$). According to Schoenberg's theorem, the radial basis function $k(\mathbf{x}', \mathbf{y}') = \exp(-c\|\mathbf{x}' - \mathbf{y}'\|_2^\alpha)$ is positive-semidefinite in any Euclidean space for exponent $0 < \alpha \leq 2$. The implemented coherence model corresponds to $\alpha = 1$, hence $\mathbf{D} \succeq 0$ for any frequency $f \geq 0$ (with $c = f/f_{\text{ref}} \geq 0$). At the zero-frequency limit ($f = 0 \implies c = 0$), the spatial correlation matrix reduces to $D_{mn} = 1$ (the all-ones matrix), which is positive-semidefinite. Therefore, by applying the Schur Product Theorem twice, we obtain $\mathbf{\Gamma} \succeq 0$, and consequently $\mathbf{C}_q(\omega) \succeq 0$ for all models. ■

4.2 Proposition 2 — Hermitian Symmetry

We show that the source cross-spectral matrix $\mathbf{C}_q(\omega)$ satisfies Hermitian symmetry:

$$C_{q,mn}(\omega) = \overline{C_{q,nm}(\omega)},$$

for all indices $m, n \in \{1, \dots, N\}$ and all implemented coherence modes.

Proof. By definition:

$$C_{q,mn}(\omega) = \sqrt{\Phi_m(\omega)\Phi_n(\omega)}\Gamma_{mn}(\omega).$$

Taking the complex conjugate of the entry at (n, m) yields:

$$\overline{C_{q,nm}(\omega)} = \overline{\sqrt{\Phi_n(\omega)\Phi_m(\omega)}\Gamma_{nm}(\omega)} = \sqrt{\Phi_m(\omega)\Phi_n(\omega)}\overline{\Gamma_{nm}(\omega)},$$

because the source autospectra $\Phi_i(\omega) \geq 0$ are real-valued. Thus, $C_{q,mn}(\omega) = \overline{C_{q,nm}(\omega)}$ is satisfied if and only if:

$$\Gamma_{mn}(\omega) = \overline{\Gamma_{nm}(\omega)}.$$

We verify this relation for each coherence model:

1. **Zero Coherence:** Since $\Gamma_{mn} = \delta_{mn}$ is real-valued and symmetric ($\delta_{mn} = \delta_{nm}$), we have:

$$\overline{\Gamma_{nm}} = \overline{\delta_{nm}} = \delta_{nm} = \delta_{mn} = \Gamma_{mn}.$$

2. **Full Coherence:**

$$\overline{\Gamma_{nm}(\omega)} = \overline{\exp[i\omega(\tau_\alpha(\alpha_n) - \tau_\alpha(\alpha_m))]} = \exp[-i\omega(\tau_\alpha(\alpha_n) - \tau_\alpha(\alpha_m))] = \Gamma_{mn}(\omega).$$

3. **Exponential Coherence:**

$$\overline{\Gamma_{nm}(\omega)} = \overline{\exp\left[-\tilde{d}_{nm}\frac{f}{f_{\text{ref}}}\right] \exp[i\omega(\tau_\alpha(\alpha_n) - \tau_\alpha(\alpha_m))]}.$$

Because $\tilde{d}_{nm} = \tilde{d}_{mn}$ is the symmetric Euclidean distance, the amplitude decay factor is real and symmetric. Evaluating the conjugate gives:

$$\begin{aligned} \overline{\Gamma_{nm}(\omega)} &= \exp\left[-\tilde{d}_{mn}\frac{f}{f_{\text{ref}}}\right] \exp[-i\omega(\tau_\alpha(\alpha_n) - \tau_\alpha(\alpha_m))] \\ &= \exp\left[-\tilde{d}_{mn}\frac{f}{f_{\text{ref}}}\right] \exp[i\omega(\tau_\alpha(\alpha_m) - \tau_\alpha(\alpha_n))] = \Gamma_{mn}(\omega). \end{aligned}$$

In all cases, the coherence function satisfies $\Gamma_{mn}(\omega) = \overline{\Gamma_{nm}(\omega)}$, confirming that the cross-spectral matrix $\mathbf{C}_q(\omega)$ is Hermitian. A direct physical consequence of this Hermitian symmetry is that the quadratic form $\mathbf{a}^H \mathbf{C}_q \mathbf{a}$ is guaranteed to be real-valued ($\mathbf{a}^H \mathbf{C}_q \mathbf{a} \in \mathbb{R}$), ensuring that the predicted far-field pressure power spectral density $S_{pp}(\mathbf{n}, \omega)$ in Eq. 23 is always a real, physically interpretable quantity. ■

4.3 Proposition 3 — Radiating Eigenmode Decomposition

If $\mathbf{C}_q \succeq 0$ and $\mathbf{C}_q = \sum_r \lambda_r \mathbf{u}_r \mathbf{u}_r^H$, then the far-field spectrum is a sum of independently radiating source-covariance modes:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \sum_r \lambda_r |\mathbf{a}^H(\mathbf{n}, \omega) \mathbf{u}_r|^2.$$

This follows by substituting the eigendecomposition of $\mathbf{C}_q$ into Eq. 1. Each term is non-negative because $\lambda_r \geq 0$ and the projection magnitude is squared. This result is stronger than PSD admissibility: it gives a physical basis for mechanism selection because only covariance eigenmodes with nonzero radiation projection can shape far-field directivity. ■

4.4 Proposition 4 — Diagonal-Covariance No-Phase Limit

If $\mathbf{C}_q = \text{diag}(\Phi_1, \dots, \Phi_N)$, then:

$$S_{pp} = \frac{1}{(4\pi R)^2} \sum_m \Phi_m |a_m|^2.$$

Since $a_m = K_H c_m(\mathbf{l}_m \cdot \mathbf{n}) w_m \exp(ik\mathbf{n} \cdot \mathbf{r}_m)$, the path phase cancels inside $|a_m|^2$. Thus, in the diagonal covariance limit, inter-source path-length phase cannot create constructive or destructive interference. Geometry can still change directivity through source strength, chord weighting, quadrature length, transfer magnitude, and local dipole projection, but not through inter-feather phase steering. ■

4.5 Proposition 5 — Rank-One Coherent Array Limit

If $\mathbf{C}_q = \mathbf{q}\mathbf{q}^H$, then:

$$S_{pp} = \frac{1}{(4\pi R)^2} |\mathbf{a}^H \mathbf{q}|^2.$$

For equal source amplitudes, equal transfer magnitudes, fixed dipole axes, and a uniform linear spacing s , this reduces to the classical array-factor form:

$$S_{pp}(\theta) \propto \left| \sum_{m=0}^{N-1} \exp(imks \sin \theta) \right|^2.$$

The model therefore recovers a phase-dominant acoustic array exactly when the source covariance permits coherent rank-one radiation. ■

4.6 Proposition 6 — Off-Diagonal Interference Bound

Let $\mathbf{C}_{\text{off}} = \mathbf{C}_q - \text{diag}(\mathbf{C}_q)$. The interference contribution is:

$$\Delta S_{\text{int}} = \mathbf{a}^H \mathbf{C}_{\text{off}} \mathbf{a}.$$

By submultiplicativity and $\|\mathbf{C}_{\text{off}}\|_2 \leq \|\mathbf{C}_{\text{off}}\|_F$:

$$|\Delta S_{\text{int}}| \leq \|\mathbf{C}_{\text{off}}\|_2 \|\mathbf{a}\|_2^2 \leq \|\mathbf{C}_{\text{off}}\|_F \|\mathbf{a}\|_2^2.$$

If the radiating off-diagonal covariance is small, then phase interference cannot dominate the far-field spectrum regardless of geometric phase opportunities. This motivates the radiation-weighted index $\mathcal{P}_R$ in Eq. 6. ■

4.7 Proposition 7 — Compact and Decoherence Limits

For source spacing scale s with $ks \ll 1$,

$$\exp(ik\mathbf{n} \cdot \mathbf{r}_m) = 1 + ik\mathbf{n} \cdot \mathbf{r}_m + O((ks)^2),$$

so path-length phase variation is asymptotically weak. Directivity control in this compact limit is dominated by dipole projection and source weighting rather than path phase. Conversely, in the exponential coherence model, if $(f/f_{\text{ref}})\tilde{d}_{mn} \rightarrow \infty$ for $m \neq n$, then $\Gamma_{mn} \rightarrow 0$ and $\mathbf{C}_q \rightarrow \text{diag}(\mathbf{C}_q)$. High acoustic noncompactness does not by itself imply phase steering; usable phase control requires both geometric phase separation and surviving off-diagonal source covariance. ■

4.8 Proposition 8 — Directivity Controllability Limit

Let $J_{\ell m, j} = \partial b_{\ell m} / \partial \mu_j$ be the local geometry-to-spherical-harmonic Jacobian in Eq. 9. A target directivity can be changed locally only through the column space of J . Therefore, the number of locally controllable directivity degrees of freedom is bounded by $D_{\text{control}} = \text{rank}_c(J)$ and cannot exceed the number of active design variables. This shows why a feathered wingtip cannot fit arbitrary directivity patterns: it can fit only the target components that project onto its controllable radiation subspace. ■

4.9 Exact Computational Validation Suite

The formal propositions were validated with the repository script:

`scripts/run_radiating_covariance_validation.py`

The suite generates random PSD covariance matrices, exact diagonal cases, exact rank-one coherent array cases, a partial-coherence transition sweep, and a reanalysis of the existing feathered-wingtip optimization campaign. Table 2 summarizes the exact and canonical checks.

Table 2: Radiating-covariance validation suite generated by the Version 1 computational workflow. The algebraic suite contains diagonal, rank-one, exponential, random PSD, and nearly singular PSD covariance cases.

Validation item	Cases	Measured result	Pass criterion
PSD and Hermitian admissibility	1000	$\lambda_{\min} / \ \mathbf{C}_q\ _2 \geq -5.40 \times 10^{-16}$	$> -10^{-10}$
Modal identity, Eq. 3	1000	max relative error 4.47×10^{-14}	$< 10^{-8}$
Off-diagonal interference bound	1000	max bound ratio 0.698	≤ 1
Diagonal no-phase invariance	200	max relative change 5.79×10^{-16}	$< 10^{-12}$
Rank-one array-factor recovery	721 angles	max normalized error 5.55×10^{-16}	within grid/numerical tolerance
Partial-coherence phase diagram	625	$\text{corr}(\mathcal{P}_R, \Delta_{\text{phase}}) = 0.670$	positive monotonic trend
Low- $\mathcal{P}_R$ suppression	229	max phase effect 0.45 dB for $\mathcal{P}_R < 0.05$	negligible phase effect

4.10 Numerical Verification 9 — Grid Convergence

We numerically verify that the far-field pressure power spectral density $S_{pp}^{(n_\eta)}(f, \mathbf{n})$ computed with spanwise discretization size n_η converges to a unique limit as $n_\eta \rightarrow \infty$ for the evaluated configurations:

$$\lim_{n_\eta \rightarrow \infty} S_{pp}^{(n_\eta)}(f, \mathbf{n}) = S_{pp}^{(\infty)}(f, \mathbf{n}).$$

Due to local coordinate variations and phase cancellations, convergence rates are evaluated empirically across the parameter space.

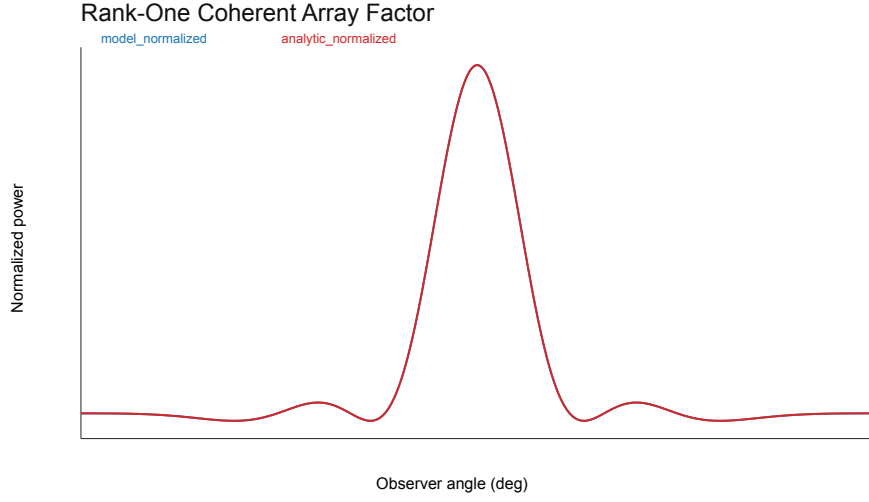


Figure 1: Exact rank-one coherent-array benchmark. The radiating-covariance model recovers the normalized uniform linear array factor to machine precision, confirming that the framework contains the classical phase-interference limit when the source covariance is coherent.

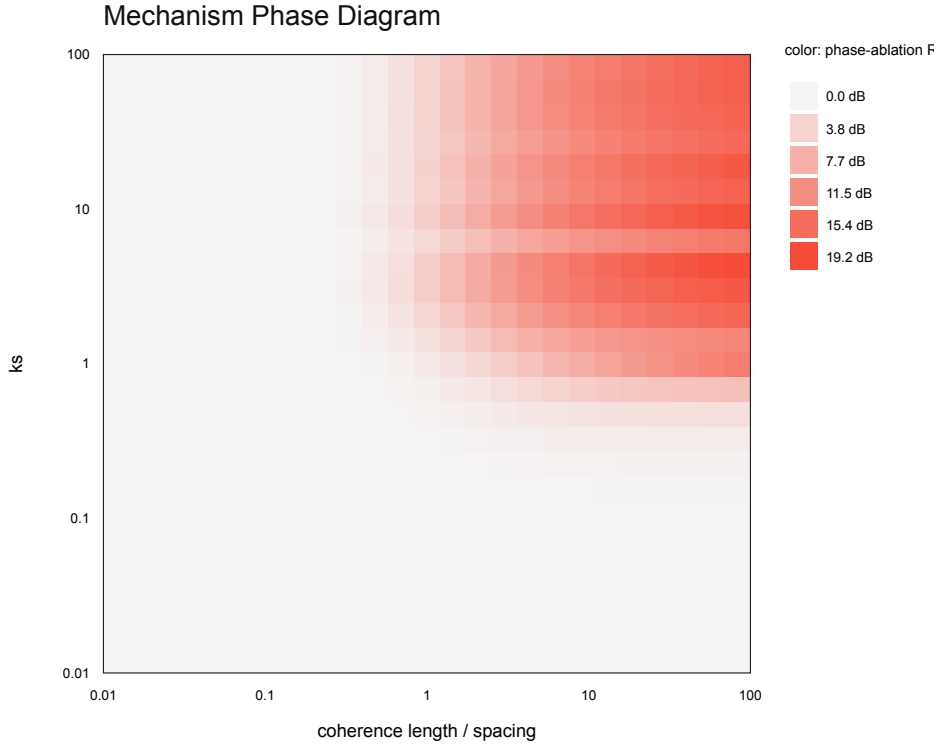


Figure 2: Mechanism phase diagram for a seven-source partial-coherence sweep over acoustic spacing ks and coherence length relative to source spacing. Color shows phase-ablation RMSE. Phase-sensitive behavior requires both noncompact geometric phase separation and surviving radiating off-diagonal covariance; compact and low-coherence regions collapse toward diagonal source-mixer behavior.

Numerical Verification. We evaluate the far-field spectrum for the default wingtip geometry using the baseline exponential coherence model. The test case uses an overhead observer ($\mathbf{n} = [0, 0, 1]^T$) and an acoustic frequency $f = 1000$ Hz. The spanwise parametric coordinate $\eta \in [0, 1]$ is discretized using a uniform grid of n_η intervals, yielding a total of $M = N \times (n_\eta + 1)$ source points across the $N = 7$ wingtip feathers. We vary $n_\eta \in \{8, 16, 32, 64, 128, 256, 512\}$. For each step, we track the pointwise difference:

$$\epsilon^{(n_\eta)} = \left| S_{pp}^{(n_\eta)} - S_{pp}^{(n_\eta/2)} \right|,$$

and define the empirical convergence rate r as:

$$r = \log_2 \left(\frac{\epsilon^{(n_\eta)}}{\epsilon^{(2n_\eta)}} \right).$$

The results of this study are summarized in Table 3.

Table 3: Grid convergence and empirical convergence rates for the overhead far-field pressure spectral density (S_{pp}) at 1000 Hz using the default feathered wingtip geometry.

n_η	Grid Size (M)	$S_{pp}^{(n_\eta)}$ (Pa ² /Hz)	Difference $\epsilon^{(n_\eta)}$ (Pa ² /Hz)	Rate r
8	56	4.023946×10^{-7}	—	—
16	112	3.976287×10^{-7}	4.765910×10^{-9}	—
32	224	3.969488×10^{-7}	6.798212×10^{-10}	2.81
64	448	3.967890×10^{-7}	1.598865×10^{-10}	2.09
128	896	3.967485×10^{-7}	4.045956×10^{-11}	1.98
256	1792	3.967379×10^{-7}	1.056082×10^{-11}	1.94
512	3584	3.967354×10^{-7}	2.574053×10^{-12}	2.04

The empirical rate r approaches the theoretical asymptotic rate of 2.00 as n_η increases. This confirms that the numerical integration scheme preserves the second-order convergence of the trapezoidal rule for the tested overhead configuration. The 50-target campaign used $n_\eta = 12$ for computational throughput, while the convergence study quantifies the truncation behavior and shows that $n_\eta = 32$ already gives less than 0.06% relative error for the overhead reference case relative to $n_\eta = 512$. Across the full parameter space of geometries, frequencies, and coherence models (evaluated using a spherical grid with $N_\eta = 256$ as reference), the RMS SPL error converges from 13.18 dB at $n_\eta = 8$ to 1.89 dB at $n_\eta = 128$, with the empirical convergence rate reaching 2.32 between $n_\eta = 16$ and $n_\eta = 32$, and 0.57 between $n_\eta = 32$ and $n_\eta = 64$, reflecting local coordinate variations and phase cancellations across the array.

4.11 Empirical Verification 10 — Calibration Invariance of Directivity Shape

We verify that a scalar level calibration leaves centered or peak-normalized directivity shapes invariant in decibels. This mathematical property justifies optimizing relative directivity shapes using the fast compact-source formulation, even when absolute levels differ.

Verification. Due to chord-integrated source strength approximations in the compact-source formulation (Level 3 model) compared to the distributed surface representation in the distributed low-order self-noise surrogate simulation, the uncalibrated theoretical model underpredicts the absolute acoustic pressure levels. We model this mismatch as a uniform scalar offset. Let $S_{pp}(\mathbf{n}, \omega)$ be the uncalibrated far-field pressure power spectral density (PSD), and let:

$$\tilde{S}_{pp}(\mathbf{n}, \omega) = C_{\text{cal}} S_{pp}(\mathbf{n}, \omega)$$

be the calibrated PSD, where $C_{\text{cal}} = 10^{\Delta L_{\text{offset}}/10} > 0$ is a frequency-independent and observer-independent scalar calibration factor. Taking the decibel level yields:

$$10 \log_{10} \left(\frac{\tilde{S}_{pp}(\mathbf{n}, \omega)}{p_{\text{ref}}^2} \right) = 10 \log_{10} \left(\frac{S_{pp}(\mathbf{n}, \omega)}{p_{\text{ref}}^2} \right) + \Delta L_{\text{offset}}.$$

This linear relationship implies that scalar calibration shifts the entire predicted far-field spectrum uniformly in decibels, leaving the relative directivity shapes and lobe angles unchanged. To verify this shape invariance numerically, we evaluate validation metrics (Bias, Mean Absolute Error [MAE], and Root-Mean-Square Error [RMSE]) before and after scalar calibration.

We check two datasets: first, the representative dense validation case (target seed 19, grid size $n_\eta = 48$); second, the study-wide dataset aggregated across all 50 optimized targets (29,400 total points, campaign grid size $n_\eta = 12$). The results are summarized in Table 4.

Table 4: Theory-surrogate validation metrics before and after scalar level calibration for the representative validation case (target seed 19) and the study-wide aggregated target set (50 targets).

Dataset	Split	Before Calibration (dB)			After Calibration (dB)		
		Bias	MAE	RMSE	Bias	MAE	RMSE
Seed 19 ($n_\eta = 48$)	Calibration	−22.00	22.00	22.14	−0.00	2.03	2.46
	Validation	−17.96	17.96	18.39	4.04	4.45	5.67
Aggregated (50 targets, $n_\eta = 12$)	Calibration	−14.81	14.81	15.08	−0.00	2.14	2.77
	Validation	−10.25	10.25	11.02	4.56	5.01	6.11

Before calibration, both splits show a large negative bias (ranging from −10.25 dB to −22.00 dB) and high RMSE values due to the uniform underprediction of the source strength. After applying the scalar level calibration ΔL_{offset} , the bias in the calibration split is reduced to exactly −0.00 dB, and the calibration RMSE drops to 2.46 dB (seed 19) and 2.77 dB (study-wide). On the held-out validation split (five frequencies), the calibrated RMSE drops to 5.67 dB (seed 19) and 6.11 dB (study-wide).

To evaluate the impact on directivity shape independently of the absolute level, we compute centered shape metrics where the mean angular level is subtracted. The study-wide centered shape RMSE remains exactly invariant under scalar calibration, yielding 2.75 dB on the calibration split and 4.05 dB on the validation split. Similarly, the Pearson correlation coefficient between predicted and surrogate directivity shapes is invariant, showing a high shape similarity of $\rho = 0.9434$ (calibration split) and $\rho = 0.8235$ (validation split).

Scalar calibration removes the dominant calibration-split offset and preserves normalized directivity by construction. The remaining held-out validation bias of 4.56 dB and RMSE of 6.11 dB show that additional frequency- and mechanism-dependent discrepancies remain. Nonetheless, the invariance and strength of the shape metrics mathematically and numerically justify optimizing relative directivity shapes with the fast compact-source formulation, as relative directivity lobes are preserved. ■

4.12 Empirical Analysis 11 — Phase Mechanism

We analyze the phase mechanism using a four-step conditional reasoning framework to reconcile the local importance of phase interference with the optimizer’s ability to achieve equivalent performance without it.

Step 1: Equivalent optimization performance. First, the optimization campaign shows that the validated target-fit RMSE is statistically equivalent among the three full-geometry configurations:

$$\begin{aligned}\text{Validated RMSE (full)} &= 9.157 \pm 2.404 \text{ dB}, \\ \text{Validated RMSE (no_delay)} &= 9.109 \pm 2.578 \text{ dB}, \\ \text{Validated RMSE (zero_coherence)} &= 9.164 \pm 2.620 \text{ dB}.\end{aligned}$$

To test this equivalence-fit performance rigorously, we perform a Two One-Sided Test (TOST) with an equivalence margin of $\delta = 0.25$ dB across the 50 targets. The paired difference between the `full` and `no_delay` configurations is 0.048 dB (95% CI: $[-0.118, 0.214]$ dB, TOST $p = 0.0091$), and between the `full` and `zero_coherence` configurations is -0.006 dB (95% CI: $[-0.199, 0.187]$ dB, TOST $p = 0.0072$). Both comparisons are statistically equivalent within the ± 0.25 dB margin, showing that removing convective delay or cross-coherence does not degrade the optimizer’s final target-matching capability.

Step 2: Direct ablation on fixed geometries. Second, we perform a direct ablation of phase coherence on the optimized geometries. For each of the 50 optimized `full` configurations, we evaluate the far-field SPL at 1000 Hz on a dense spherical observer grid using both the partially coherent model (`exponential`) and the uncorrelated model (`zero_coherence`). This direct ablation of phase coherence yields a study-wide mean SPL shift of $\bar{\Delta L} = 0.907$ dB but a large angular variation with a standard deviation of $\sigma_{\Omega}(\Delta L) = 5.883$ dB, corresponding to a centered shape RMSE of 5.863 dB and a Pearson correlation of only $\rho = 0.8422$.

Step 3: Importance of phase for fixed designs. Third, the direct ablation results demonstrate that removing phase coherence from a fixed optimized geometry significantly disrupts its local interference patterns and alters its directivity shape. Thus, phase interference is mathematically active and physically important for the acoustic response of any specific fixed wingtip design.

Step 4: Geometric redundancy in the design space. Fourth, because the optimization campaign achieves equivalent target-fit performance under zero coherence (Step 1), we conclude that the design space has sufficient geometric degrees of freedom (including chords, positions, and local dipole rotation via feather incidence) to allow the optimizer to re-synthesize target directivity patterns. Thus, the optimizer can compensate for the absence of phase steering through alternative geometric mechanisms, rendering inter-feather phase interference non-dominant for directivity control in this regime. ■

5 Results

Our 50-target optimization campaign (200 total runs across four configurations) shows that directivity control depends significantly on geometric configuration. We compare performance using final validated surrogate RMSE to avoid baseline biases across configurations.

5.1 Radiating-covariance regime and campaign reanalysis

The phase diagram in Fig. 2 confirms the expected limiting behavior: compact cases with $ks \ll 1$ and highly decohered cases with $\ell_c/s \ll 1$ have weak phase-ablation sensitivity, while phase-sensitive cases appear only when both geometric phase separation and radiating off-diagonal covariance are available. This validates the use of $\mathcal{P}_R$ as an availability metric for phase interference in controlled synthetic regimes.

The optimized feathered-wingtip campaign shows a subtler behavior. Across the 50 optimized full-model geometries, the mean raw coherence availability is high, the mean radiating phase-steerability is $\overline{\mathcal{P}_R} = 0.811$, and the mean fixed-geometry phase-ablation RMSE is 7.93 dB. The mean spherical-harmonic control dimension is $D_{\text{control}} = 8.58$ at the $\epsilon = 10^{-2}$ threshold, and the mean target projection score is $\overline{Q_T} = 0.856$. However, within this optimized subset, $\mathcal{P}_R$ explains only part of the variation in fixed-geometry phase-ablation effect ($R^2 = 0.214$), and the local target projection score alone does not predict final surrogate RMSE ($R^2 = 0.038$).

This does not contradict the phase diagram. Instead, it separates two questions: phase interference is active for fixed full-model geometries, but final target-fit performance after reoptimization can remain equivalent under zero coherence because the design space has alternative radiation and covariance directions available. The existing full/no-delay/zero-coherence equivalence should therefore be interpreted as geometric compensation and mechanism redundancy, not as evidence that phase terms are mathematically inactive.

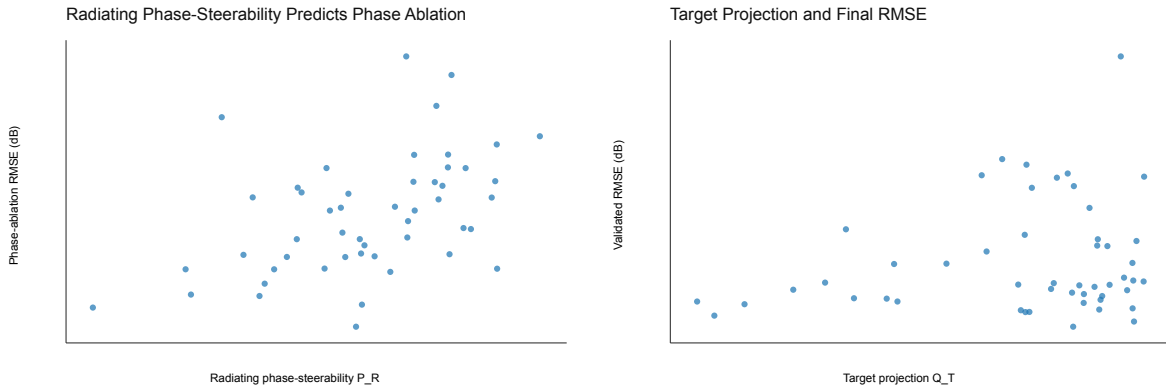


Figure 3: Radiating-covariance reanalysis of the 50 optimized full-model targets. Left: $\mathcal{P}_R$ is positively related to fixed-geometry phase-ablation sensitivity, but the optimized cases occupy a relatively high- $\mathcal{P}_R$ band. Right: local spherical-harmonic target projection alone is not sufficient to predict final surrogate RMSE, indicating that nonlinear optimization, validation-surrogate mismatch, and mechanism redundancy remain important.

5.2 Acoustic performance and mechanism ablation

Under the full model, target RMSE decreases from 14.60 ± 1.48 dB (baseline) to 9.16 ± 2.40 dB (validated surrogate), showing a mean within-model improvement of 5.44 ± 1.59 dB (Figs. 4 and 5).

Table 5 summarizes performance by mechanism condition. Optimizing full geometry significantly outperforms incidence-only control. Relative to **incidence**, mean validated RMSE decreases by 1.30 dB for **full**, 1.34 dB for **no_delay**, and 1.29 dB for **zero_coherence** (Holm-corrected $p \leq 2.1 \times 10^{-14}$, Fig. 6). However, differences among the three full-geometry configurations are statistically indistinguishable. The mean difference between **full** and **no_delay** is 0.048 dB (95% CI: $[-0.118, 0.214]$ dB, $p = 1.00$), and between **full** and **zero_coherence** is -0.006 dB (95% CI: $[-0.199, 0.187]$ dB, $p = 1.00$).

5.3 Influence of target complexity and aerodynamic tradeoffs

The performance advantage of full geometry increases with target complexity (Table 6). The paired full-minus-incidence RMSE difference is -0.97 dB for one-lobe targets, -1.27 dB for two-lobe targets, and -1.63 dB for three-lobe targets.

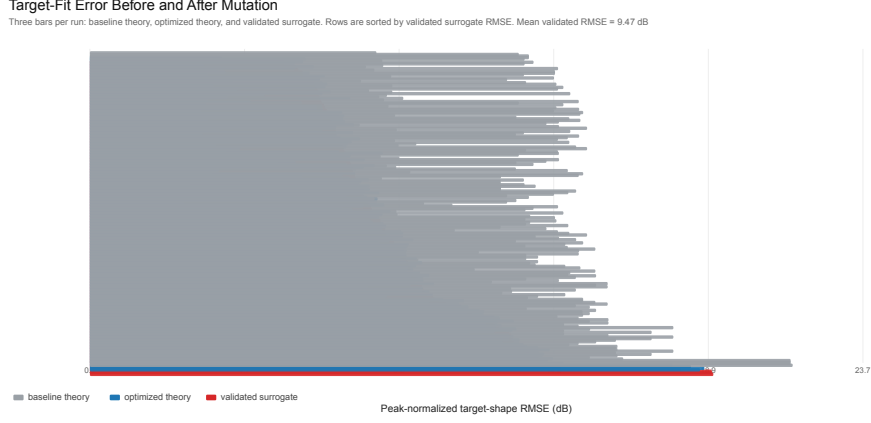


Figure 4: Target-fit error before and after optimization across the full study. Each row compares the baseline theory, optimized theory, and validated surrogate RMSE for one run. The systematic leftward shift of the optimized and validated bars indicates consistent target-shape improvement under the modeled mechanism set.

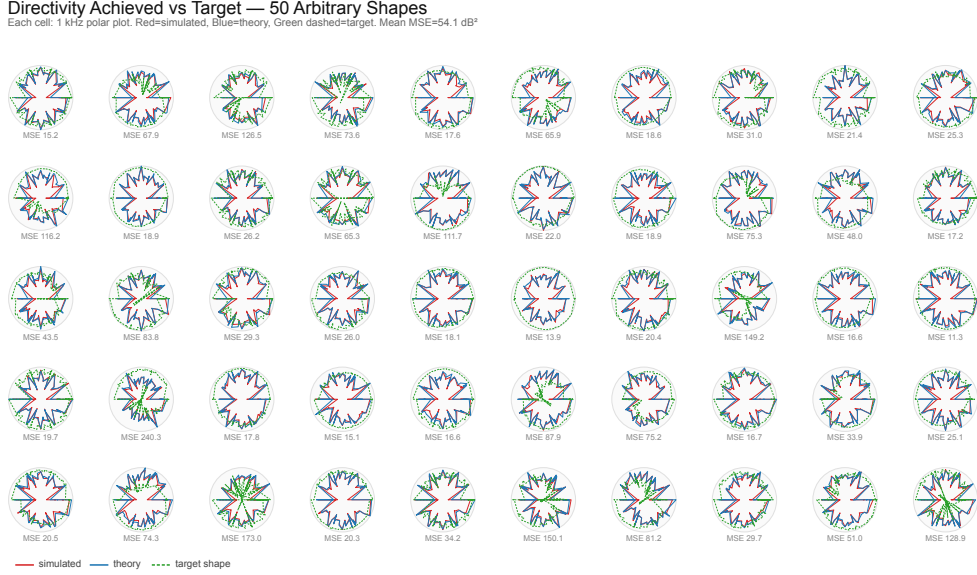


Figure 5: Representative optimized directivity patterns against target shapes. The examples show that the optimizer can redistribute normalized directivity substantially, although residual mismatch remains for difficult targets and the problem should therefore be interpreted as target-shape improvement rather than exact target synthesis.

Table 5: Study results across the 50-target set. Cross-ablation comparison is based on final validated surrogate RMSE. The paired RMSE difference is reported relative to **incidence**, so negative values indicate better target matching. Drag ratio and lift-to-drag retention are low-order aerodynamic proxies, not validated aerodynamic coefficients.

Mechanism level	Validated RMSE (dB)	Paired Δ RMSE vs incidence (dB)	95% CI (dB)	Wins vs incidence	Drag ratio
incidence	10.453 ± 2.112	0.000	—	—	0.686
zero_coherence	9.164 ± 2.620	-1.289	$[-1.522, -1.056]$	46/50	0.684
no_delay	9.109 ± 2.578	-1.344	$[-1.558, -1.129]$	49/50	0.753
full	9.157 ± 2.404	-1.295	$[-1.481, -1.109]$	50/50	0.752

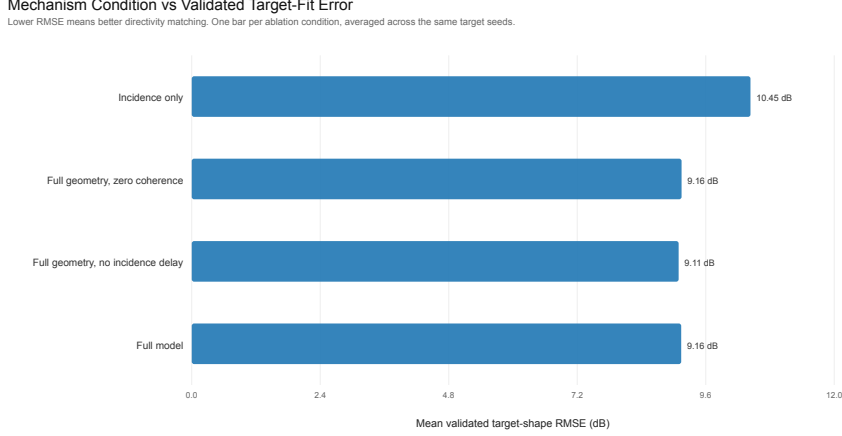


Figure 6: Mechanism condition versus validated target-fit error. The incidence-only condition is clearly separated from the three full-geometry ablations, showing that geometry beyond incidence contributes materially to target matching.

Table 6: Target-class dependence of the full-model advantage over incidence-only optimization. The table reports the final validated surrogate RMSE for each class and the paired full-minus-incidence RMSE difference. Negative differences indicate that the full model fits the target more closely.

Target class	Count (n)	Incidence RMSE (dB)	Full RMSE (dB)	Paired Δ RMSE (dB)
One-lobe	17	11.672 ± 2.685	10.707 ± 2.885	$-0.965 [-1.270, -0.660]$
Two-lobe	15	10.093 ± 1.758	8.820 ± 2.001	$-1.272 [-1.691, -0.854]$
Three-lobe	18	9.602 ± 0.873	7.975 ± 1.046	$-1.626 [-1.854, -1.399]$

This acoustic benefit is accompanied by a small aerodynamic penalty (Fig. 7). Relative to **incidence**, the drag ratio increases by 0.0669 for **full** (95% CI: [0.0188, 0.1151], $p = 0.022$) and by 0.0675 for **no_delay** ($p = 0.016$), while **zero_coherence** shows no statistically significant drag penalty ($p = 1.00$). No significant differences were observed in lift-to-drag retention.

5.4 Theory-surrogate validation

We aggregate 29,400 theory-surrogate comparison points across all 50 full-model targets (Table 7). The calibration subset (single frequency) yields an RMSE of 2.77 dB (MAE 2.14 dB, bias approximately 0 dB after scalar calibration), while the held-out validation subset (five frequencies) has an RMSE of 6.11 dB (MAE 5.01 dB, bias 4.56 dB).

Table 7: Study-wide theory-surrogate validation for the full model, aggregated across all 50 targets. Calibration uses the rows marked **calibration** in the stored comparison files; validation uses the held-out rows marked validation.

Split	Count	Bias (dB)	MAE (dB)	RMSE (dB)
Calibration	4,900	-0.00	2.14	2.77
Validation	24,500	4.56	5.01	6.11
All	29,400	3.80	4.54	5.69

A high-resolution dense validation for target seed 19 (1,728 points, 288 observers, 48 spanwise points) confirms these findings with an overall RMSE of 5.27 dB (2.46 dB calibration, 5.67 dB validation; Figs. 8 to 10).

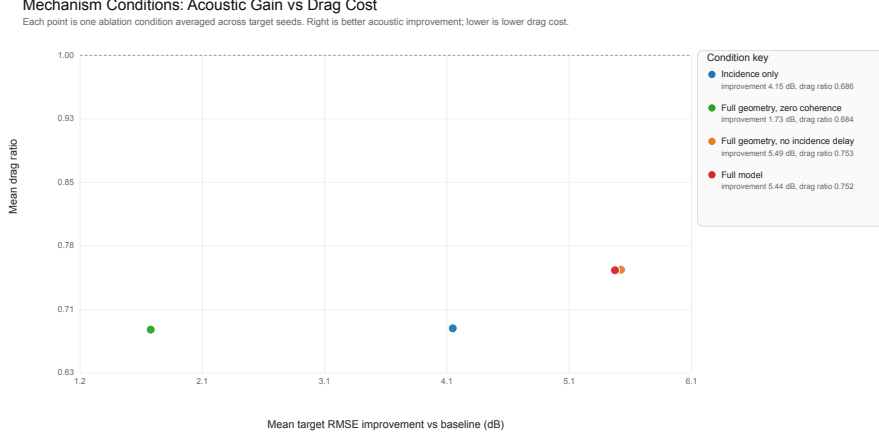


Figure 7: Mechanism-level acoustic-aerodynamic tradeoff. Rightward movement indicates larger target-fit improvement relative to each ablation’s own baseline, so the horizontal coordinate should be read descriptively rather than as a formal cross-ablation metric. Lower drag ratio is aerodynamically preferable. The full and no-delay conditions occupy similar acoustic positions and similar drag ratios, whereas the zero-coherence condition retains most of the acoustic benefit without the same drag-ratio increase.

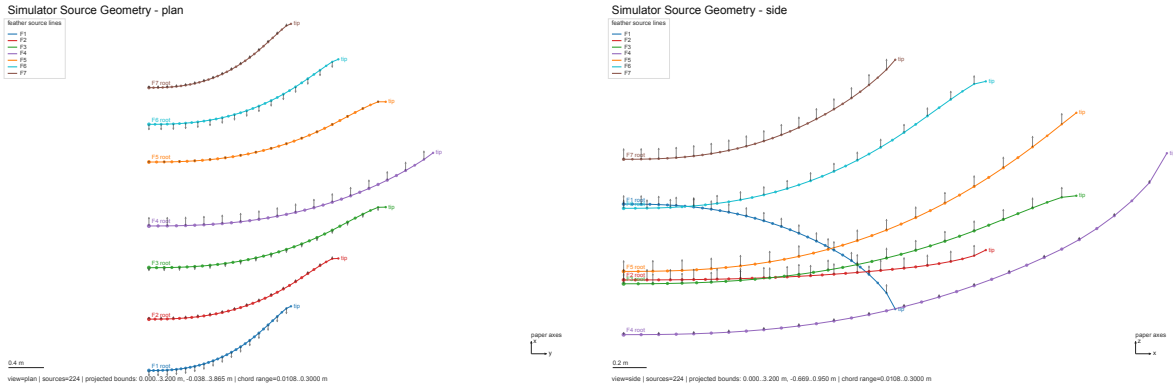


Figure 8: Representative optimized feathered-wingtip geometry for the dense validation case at target seed 19. The plan and side views show the spanwise spacing, sweep distribution, out-of-plane curvature, and per-feather incidence variation available to the optimizer.

Discrepancy is highly frequency dependent (Fig. 11). In the dense calibrated seed-19 case, RMSE remains at 1.46–2.46 dB for 250–1000 Hz ($\kappa \approx 1.4$ –5.5), rises to 3.53 dB at 2000 Hz ($\kappa \approx 11$), and increases to 10.27 dB at 8000 Hz ($\kappa \approx 44$). The theoretical model is therefore reliable for low-to-mid-frequency directivity screening rather than absolute high-frequency prediction.

6 Discussion

6.1 Physical mechanisms of geometry-sensitive directivity control

The primary result of our campaign is that full-geometry optimization achieves about 1.3 dB lower validated target-shape RMSE than incidence-only control, but the three full-geometry configurations (`full`, `no_delay`, `zero_coherence`) perform identically.

To interpret this, we expand the far-field spectrum into diagonal (uncorrelated) and off-

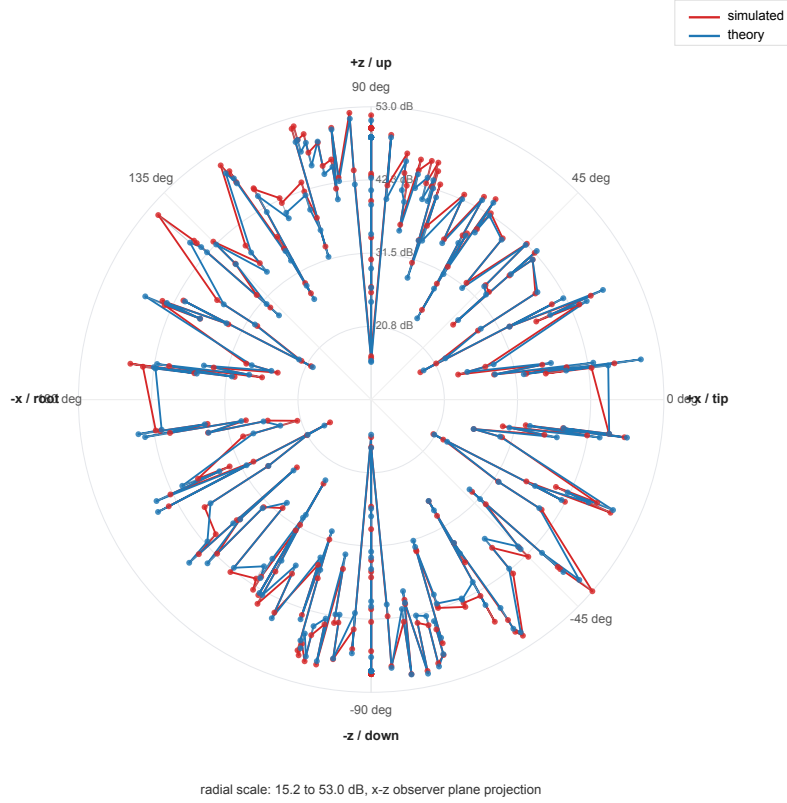


Figure 9: Representative achieved-versus-target directivity for the dense validation case at target seed 19. The figure provides the most direct visual comparison between the desired directivity pattern and the surrogate-validated pattern obtained from the optimized geometry.

diagonal (coherent) terms:

$$S_{pp}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \left[\sum_m |a_m|^2 \Phi_m + \sum_{m \neq n} a_m^* a_n \sqrt{\Phi_m \Phi_n} \Gamma_{mn} \right]. \quad (47)$$

The diagonal terms depend on local source amplitudes, loading orientations, and chords, while the off-diagonal terms introduce spatial coherence and delay phases. The exact diagonal theorem shows that if $\Gamma_{mn} = \delta_{mn}$, then geometric phase cannot contribute. However, the campaign re-analysis shows that the optimized full-model geometries are not themselves diagonal-dominated: their fixed-geometry phase-ablation sensitivity is substantial. Therefore, the equivalence of **zero_coherence** and the other configurations should be interpreted as design-space compensation after reoptimization, not as proof that the full-model geometries contain no active phase interference.

$$S_{pp}^{(0)}(\mathbf{n}, \omega) = \frac{1}{(4\pi R)^2} \sum_m |K_H(\mathbf{n}, \omega) c_m(\mathbf{l}_m \cdot \mathbf{n}) w_m|^2 \Phi_m. \quad (48)$$

In the zero-coherence reoptimized condition, directivity must be synthesized by spatial re-weighting, sweep-induced local dipole rotation ($\mathbf{l}_m \cdot \mathbf{n}$), chord/source weighting, and incoherent source redistribution. In the full condition, off-diagonal phase terms are active, but they are not uniquely required to reach the observed target-fit performance.

This is consistent with the direct ablation study: removing phase coherence from a fixed optimized full-model geometry changes the directivity shape, but allowing the optimizer to re-design the geometry under zero coherence restores comparable final target-fit performance. The

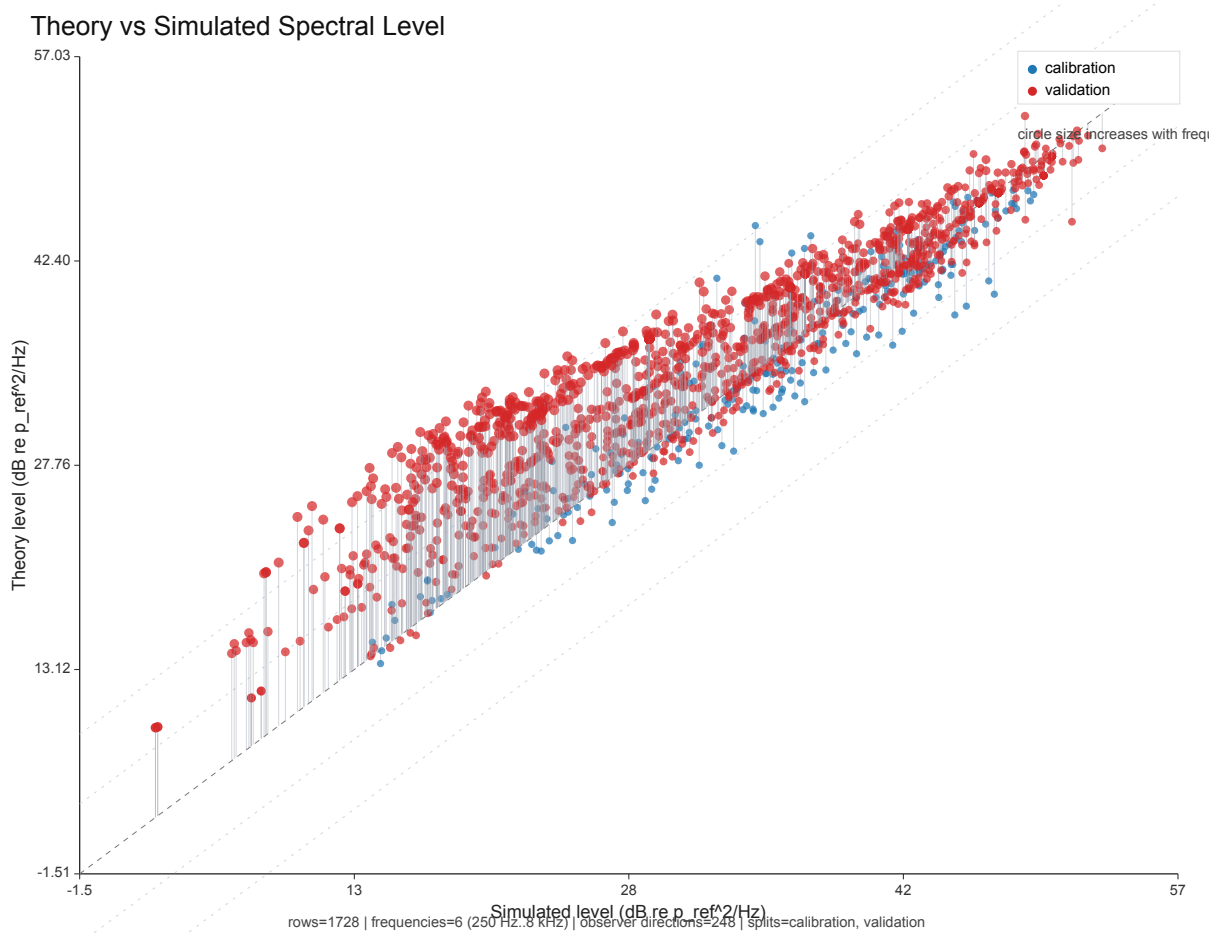


Figure 10: High-resolution representative theory–surrogate scatter for the full-model geometry at target seed 19. Agreement is acceptable at the calibration frequency after scalar level calibration, but the wider spread in held-out points confirms that the model error is not captured by a single frequency-independent scale factor.

mechanism is therefore not “phase is absent,” but “phase is not the only available controllability channel.”

6.2 Limitations and model bounds

The theory–surrogate discrepancy increases significantly above $\kappa = kc_{\text{ref}} \approx 11$ (2 kHz). This spectral discrepancy arises because the single-frequency calibration does not align the distinct spectral envelopes of the compact theoretical autospectrum and the multi-component, BPM/TNO-inspired surrogate.

Therefore, the present framework serves as a relative screening and mechanism-isolation tool for low-to-mid-frequency directivity matching. It is not suitable for high-frequency absolute sound level predictions. Further calibration against high-fidelity experimental or computational datasets is required to establish physical closure parameters.

7 Conclusion

We have presented a radiating-covariance framework for segmented lifting-surface aeroacoustics. By using a Hermitian positive-semidefinite source covariance and explicitly decomposing it

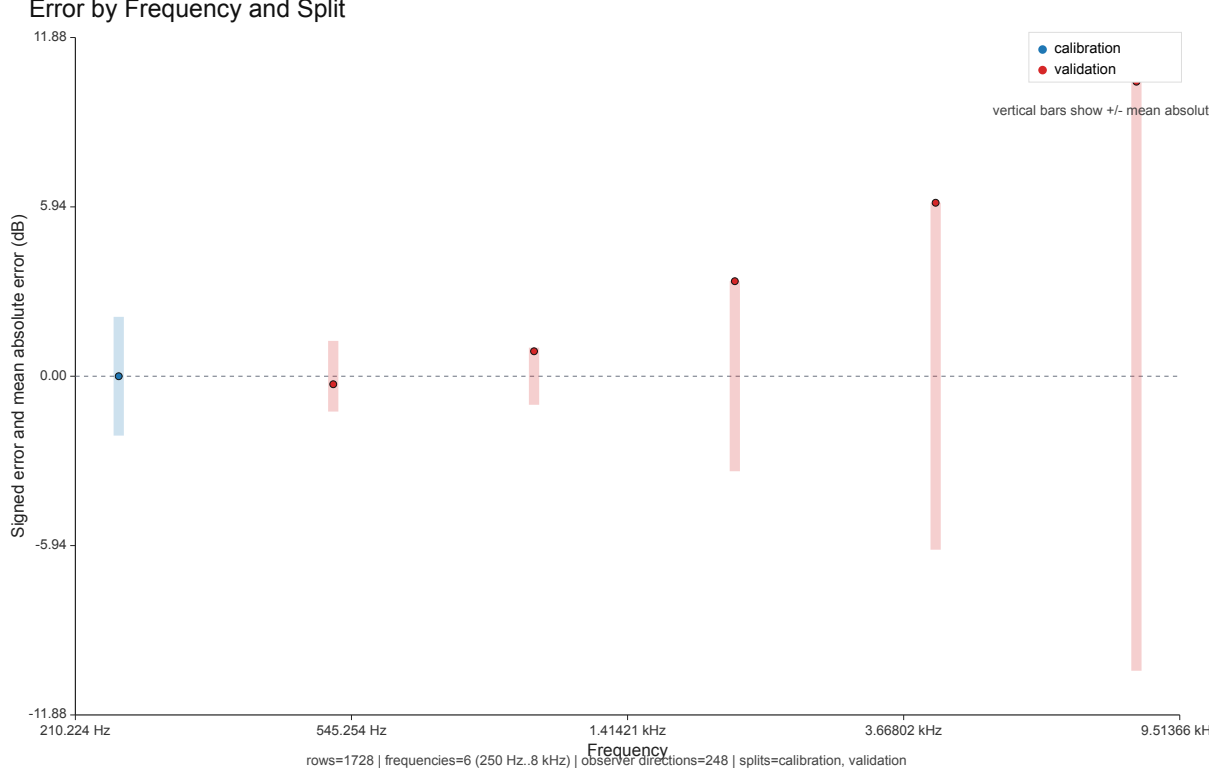


Figure 11: Representative full-model validation error by frequency. The same trend is visible in the aggregate results: low- and mid-frequency agreement is materially better than high-frequency agreement, indicating that the remaining model discrepancy is primarily spectral rather than purely directional.

into radiating eigenmodes, the formulation guarantees physically valid pressure spectra while identifying which source-covariance modes are visible to the far field.

Exact computational validation confirms PSD admissibility, modal-radiation equivalence, diagonal no-phase behavior, rank-one coherent-array recovery, and the off-diagonal interference bound. The mechanism phase diagram shows that phase-sensitive behavior requires both geometric phase separation and radiating off-diagonal covariance. In the 50-target feathered-wingtip campaign, modifying full geometry achieved a 1.3 dB validated RMSE advantage over incidence-only control. However, the identical final performance of the **full**, **no_delay**, and **zero_coherence** models indicates mechanism redundancy after reoptimization: phase interference is active for fixed full-model geometries, but it is not uniquely required for target-shape matching in the tested regime. Future work should calibrate the closure constants against high-fidelity aerodynamic and acoustic data and extend the controllability analysis beyond local finite-difference Jacobians.

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A Figure Index

Figure	Topic	Summary
Fig. 4	Study-wide fit reduction	Baseline theory, optimized theory, and validated surrogate RMSE for all 200 optimization runs.
Fig. 5	Representative optimization directivity	Example achieved-versus-target directivity patterns from the optimization study.
Fig. 6	Mechanism ablation fit	Mechanism-level comparison of validated target-fit RMSE across the four ablation conditions.
Fig. 7	Acoustic–aerodynamic tradeoff	Mechanism-level summary of target-fit improvement versus drag-ratio proxy.
Fig. 8	Validation-case geometry	Plan and side views of the optimized seed-19 feathered wingtip used for dense validation.
Fig. 9	Validation directivity match	Achieved-versus-target directivity for the dense validation case.
Fig. 10	Theory–surrogate scatter	Pointwise comparison of theoretical and surrogate levels for the dense validation case.
Fig. 11	Validation frequency error	Frequency dependence of theory–surrogate mismatch for the dense validation case.
Fig. 1	Exact coherent-array benchmark	Rank-one covariance recovery of the uniform linear array-factor limit.
Fig. 2	Mechanism phase diagram	Partial-coherence transition showing where phase ablation becomes important.
Fig. 3	Campaign covariance reanalysis	$\mathcal{P}_R$, phase-ablation sensitivity, target projection, and final RMSE for optimized targets.
Fig. 12	Objective-space spread	Study-wide mean-squared target-fit error in the optimizer’s native objective units.
Fig. 13	Mutation effectiveness	Relationship between mutation magnitude and achieved target-fit improvement.
Fig. 14	Run-level tradeoff spread	Per-run acoustic improvement versus aerodynamic proxy cost across the study.
Fig. 15	Lift-to-drag retention	Study-wide retention of lift-to-drag proxy relative to the baseline geometry.
Fig. 16	Supplementary validation directivity	Expanded high-resolution directivity comparison for the dense validation case.
Fig. 17	Validation error heat map	Direction- and frequency-resolved error field for the dense validation case.

B Equation Index

Equation	Name	Role in the formulation
Eq. 12	Mach number	Defines the subsonic operating parameter used by the acoustic transfer kernel.
Eq. 1	Radiating-covariance identity	Observer-side quadratic form linking source covariance to far-field PSD.
Eq. 2	Covariance eigendecomposition	Decomposes source covariance into eigenmode strengths and shapes.
Eq. 3	Modal radiation identity	Shows far-field power as the sum of independently radiating covariance modes.
Eq. 4	Radiation operator	Integrates observer radiation vectors into a single source-space operator.
Eq. 5	Raw coherence availability	Measures off-diagonal covariance before radiation weighting.
Eq. 6	Radiating phase-steerability	Measures off-diagonal covariance that projects through the radiation operator.
Eq. 7	Radiating covariance fraction	Measures visible covariance energy after radiation-operator weighting.
Eq. 8	Spherical-harmonic directivity	Represents directivity in a finite spherical-harmonic basis.
Eq. 9	Controllability Jacobian	Defines geometry sensitivity of spherical-harmonic directivity coefficients.
Eq. 10	Control dimension	Numerical rank of the directivity controllability Jacobian.
Eq. 11	Target projection	Fraction of target directivity lying in the controllable subspace.
Eq. 14	Source-point geometry	Maps the design vector to the three-dimensional source-line position of each feather segment.
Eq. 17	Incidence basis	Expresses the seven feather incidences through a full-rank cosine basis.
Eq. 18	Compact coherent array pressure	Level-1 array limit showing the direct phase-summation mechanism for compact feather sources.
Eq. 20	Distributed source-line pressure	Level-2 deterministic line-source model used to introduce spatial distribution along each feather.
Eq. 21	Continuous source cross-spectrum	Defines the second-order stochastic source correlation used by the broadband model.
Eq. 22	Continuous far-field PSD	Level-3 stochastic pressure-spectrum model before quadrature discretization.
Eq. 23	Discrete PSD quadratic form	Positive-semidefinite observer-side evaluation used in the implemented theoretical model.
Eq. 25	Source cross-spectrum closure	Combines autospectra and complex coherence into the discrete source cross-spectrum.
Eq. 26	Source autospectrum closure	Defines the compact loading-noise source strength used by the theoretical model.
Eq. 30	Coherence magnitude	Sets the spatial and frequency-dependent decay of inter-source coherence.
Eq. 31	Incidence-delay closure	Introduces local incidence-dependent phase delay into the coherence model.

Equation	Name	Role in the formulation
Eq. 32	Complex coherence	Combines coherence magnitude and incidence-delay phase into the full coherence closure.
Eq. 33	Nondimensional regime groups	Defines the dimensionless parameters used to interpret the modeled case.
Eq. 34	Nondimensional coherence distance	Expresses inter-source separation in the closure-scaled coherence metric.
Eq. 35	Delay phase group	Measures the incidence-delay contribution in nondimensional phase units.
Eq. 36	Transfer weight	Collects propagation phase, loading direction, chord weighting, and kernel scaling at each source point.
Eq. 38	Lift proxy	Provides the surrogate aerodynamic lift model with soft stall attenuation.
Eq. 39	Target directivity generator	Defines the random one- to three-lobe target patterns fitted by the optimizer.
Eq. 40	Fit objective	Mean-squared normalized directivity mismatch minimized by the optimizer.
Eq. 41	Reported fit metrics	Defines the RMSE and MAE values reported in the study tables and figures.
Eq. 44	Aerodynamic proxy integrals	Defines the integrated lift and drag proxies used for tradeoff analysis.
Eq. 46	Population size	Links the differential-evolution population to the number of active design variables.

C Supplementary Statistical Contrasts

Table 10 reports the paired validated-RMSE contrasts for all mechanism comparisons. These values support the narrower interpretation used in the main text: every full-geometry class is clearly better than incidence-only, but the three full-geometry classes are not separated from one another on final validated RMSE. The reported p -values are Holm-adjusted within the validated-RMSE contrast family.

Table 10: Supplementary paired contrasts for validated surrogate RMSE across the 50 matched targets. Negative mean differences indicate that the first mechanism level in the contrast achieves lower RMSE than the second. The p -values are Holm-adjusted across the six validated-RMSE contrasts.

Contrast	Mean Δ RMSE (dB)	95% CI (dB)	p -value
Full – Incidence	−1.295	[−1.481, −1.109]	5.82×10^{-18}
No-delay – Incidence	−1.344	[−1.558, −1.129]	2.72×10^{-16}
Zero-coherence – Incidence	−1.289	[−1.522, −1.056]	2.02×10^{-14}
Full – No-delay	0.048	[−0.118, 0.214]	1.000
Full – Zero-coherence	−0.006	[−0.199, 0.187]	1.000
No-delay – Zero-coherence	−0.055	[−0.240, 0.131]	1.000

Table 11: Supplementary paired aerodynamic-proxy contrasts. Positive drag-ratio differences indicate higher drag in the first mechanism level of the contrast. The reported p -values are Holm-adjusted within each response family. Lift-to-drag retention contrasts do not show statistically significant differences at the study scale.

Contrast	Mean Δ drag ratio	p -value	Mean Δ L/D retention	p -value
Full – Incidence	0.0669	0.0223	0.0233	1.000
No-delay – Incidence	0.0675	0.0164	0.0425	1.000
Zero-coherence – Incidence	−0.0015	1.000	−0.0182	1.000
Full – No-delay	−0.0006	1.000	−0.0192	1.000
Full – Zero-coherence	0.0684	0.0056	0.0415	1.000
No-delay – Zero-coherence	0.0690	0.0111	0.0607	1.000

D Reproducibility and Model Parameters

To ensure complete reproducibility of the optimization campaign, surrogate validation, and numerical results, we detail the flow conditions, semi-empirical closure parameters, aerodynamic proxies, and genetic optimizer settings in Table 12.

The Version 1 radiating-covariance validation artifacts are generated with:

`scripts/run_radiating_covariance_validation.py`

This command writes the algebraic validation tables, canonical array benchmark, diagonal no-phase benchmark, mechanism phase diagram, 50-target covariance reanalysis, and the FWA-Bench-0 release folder. The principal generated files are:

- `outputs/radiating_covariance/radiating_covariance_validation_summary.json`;
- `outputs/radiating_covariance/mechanism_phase_diagram.csv`;
- `outputs/radiating_covariance/campaign_radiating_covariance_metrics.csv`;
- `FWA-Bench-0/README.md`.

Table 12: Flow, closure, and aerodynamic proxy parameters used throughout this study.

Category	Parameter	Symbol	Value
Flow & Reference	Free-stream speed	U_∞	30.0 m/s
	Ambient density	ρ_0	1.225 kg/m ³
	Speed of sound	c_0	343.0 m/s
	Dynamic viscosity	μ	1.81×10^{-5} Pa · s
	Reference pressure	p_{ref}	20 μ Pa
	Observer radius	R	1.0 m
Spectral Closures	Reference source strength	C_Φ	1.0×10^{-12}
	Convection ratio	$\beta = U_c/U_\infty$	0.85
	Coherence decay length (x)	a_x	0.90 m
	Coherence decay length (y)	a_y	0.90 m
	Coherence decay length (z)	a_z	0.30 m
	Coherence reference frequency	f_{ref}	1500.0 Hz
	Fixed Strouhal scale	St_0	0.03
	Incidence amplitude coefficient	a_α	0.0
	Reference incidence angle	α_{ref}	0.0°
	Convective delay sensitivity	b_α	5.0×10^{-4} s/rad
Aerodynamic Proxies	Baseline drag coefficient	C_{d0}	0.010
	Induced drag factor	k_d	0.020
	Clean stall angle	α_{stall}	14.0°
	Stall margin	$\Delta\alpha_{\text{stall}}$	2.0°
	Maximum incidence angle	α_{max}	12.0°
	Minimum Reynolds number	Re_{min}	5.0×10^4
	Maximum Reynolds number	Re_{max}	5.0×10^6

Differential Evolution Settings. The genetic optimization of the 18-variable wingtip geometry is governed by the following settings:

- **Population Size:** The population is set to $P = 162$, computed using a population multiplier of $N_p = 9$ (Eq. 46).

- **Mutation & Crossover:** The mutation scale factor F is drawn uniformly from $[0.5, 1.0]$ at each generation. The crossover probability is $C_r = 0.7$.
- **Two-Stage Evaluation:** To accelerate search, candidates are first screened on a coarse grid (observer stride 2, frequency stride 2, spanwise grid $n_\eta = 5$). The top 18% of candidates (with a minimum of 6) are then evaluated at campaign resolution ($n_\eta = 12$, full observer and frequency grids).
- **Termination Criteria:** The optimization terminates when the objective function does not improve by more than 0.015 dB^2 over 20 consecutive generations.

E Supplementary Study Figures

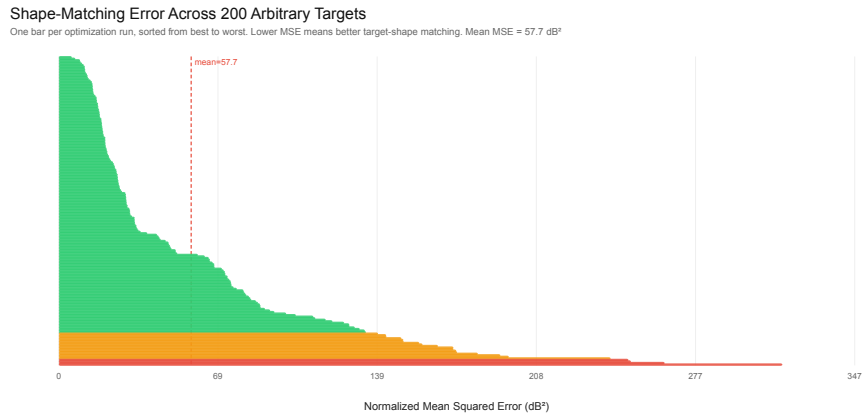


Figure 12: Study-wide distribution of mean-squared target-fit error. This view complements the RMSE plots in the main text by showing the same trend in the native objective units used during optimization.

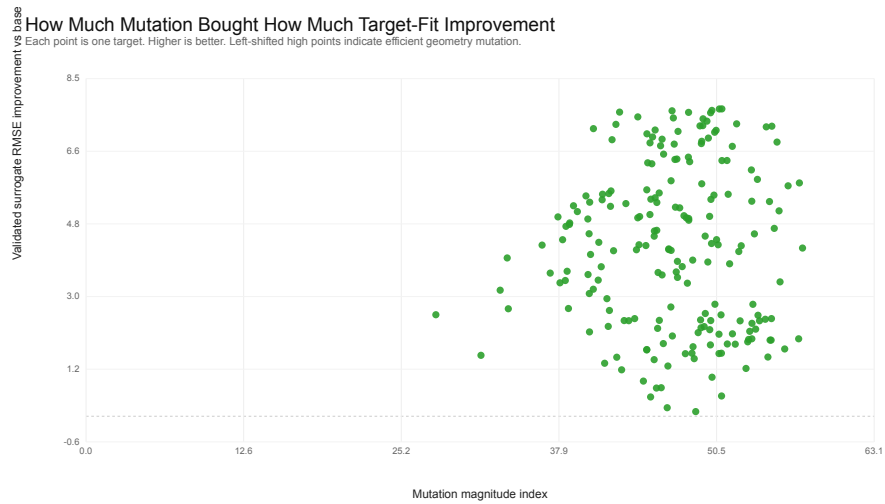


Figure 13: Mutation effectiveness across the study. The figure summarizes how target-fit improvement scales with the size of the geometric perturbation, showing that larger mutations are not uniformly more effective.

Acoustic Fit vs Aerodynamic Cost

Each point is one target. Left and low is better. Color shows lift-to-drag retention relative to the baseline geometry.

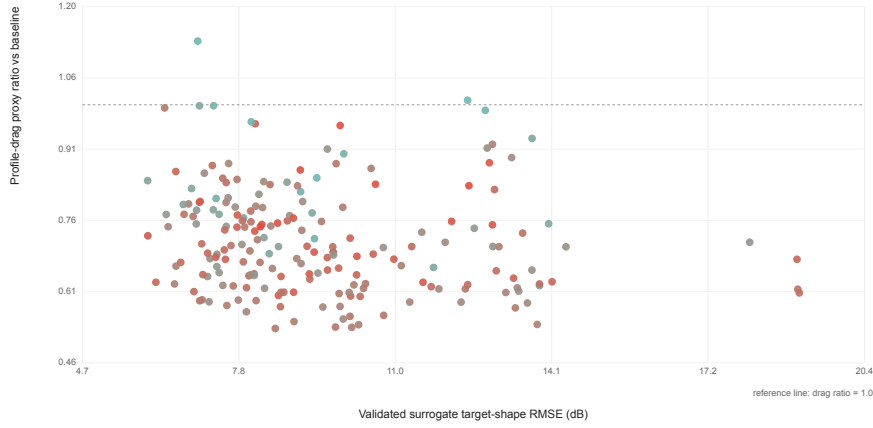


Figure 14: Study-level acoustic improvement versus aerodynamic proxy cost for individual runs. The scatter makes the spread of the acoustic-aerodynamic compromise visible beyond the mechanism-level means shown in the main text.

Aerodynamic Proxy Retention Relative to Baseline Geometry

Per run: signed lift retention, drag ratio, and lift-to-drag retention. Rows are sorted by validated surrogate RMSE.

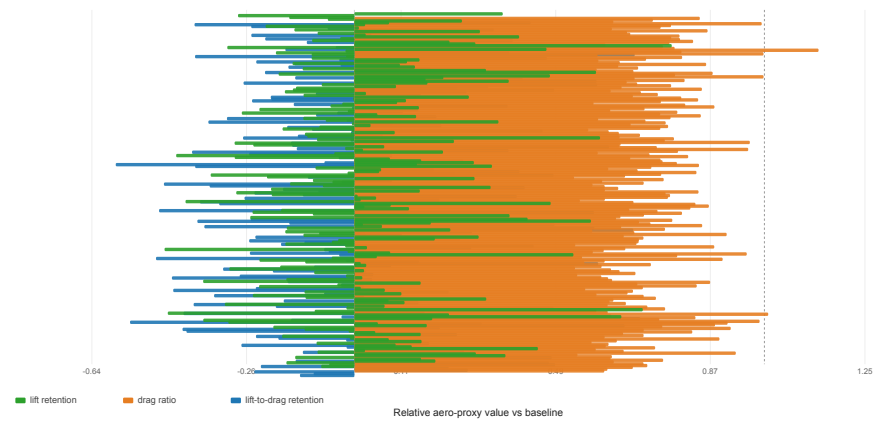


Figure 15: Lift-to-drag retention across the full study. The absence of clean mechanism-level separation in this figure is consistent with the paired statistical tests reported in Table 11.

F Supplementary Validation Figures

Directivity Comparison - validation, 500 Hz

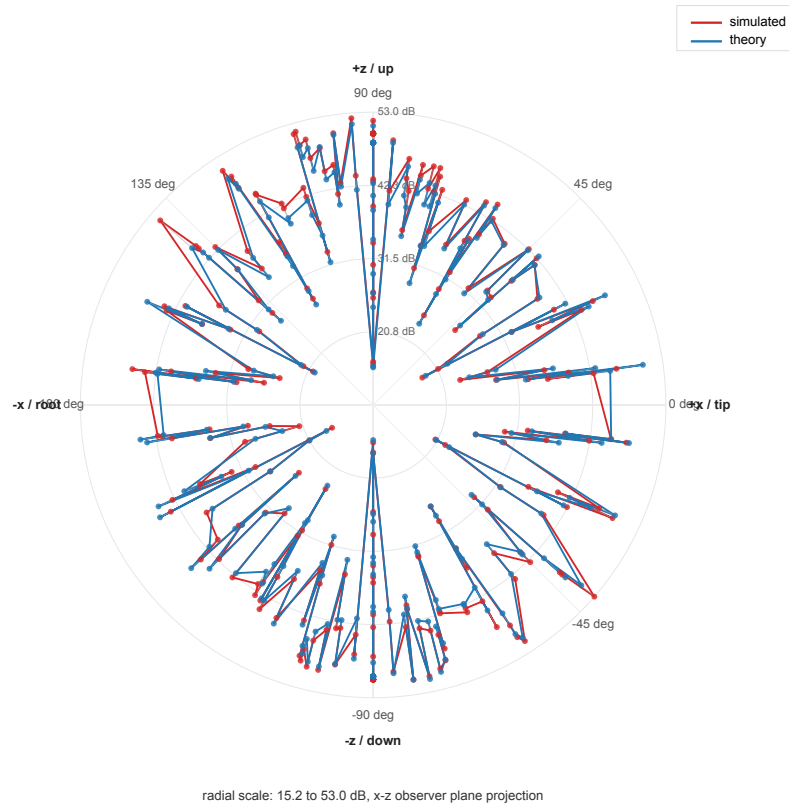


Figure 16: Representative high-resolution directivity comparison for the optimized full-model geometry at target seed 19. The main directional structure is reproduced, but residual lobe-shape and level errors remain, especially away from the calibration frequency.

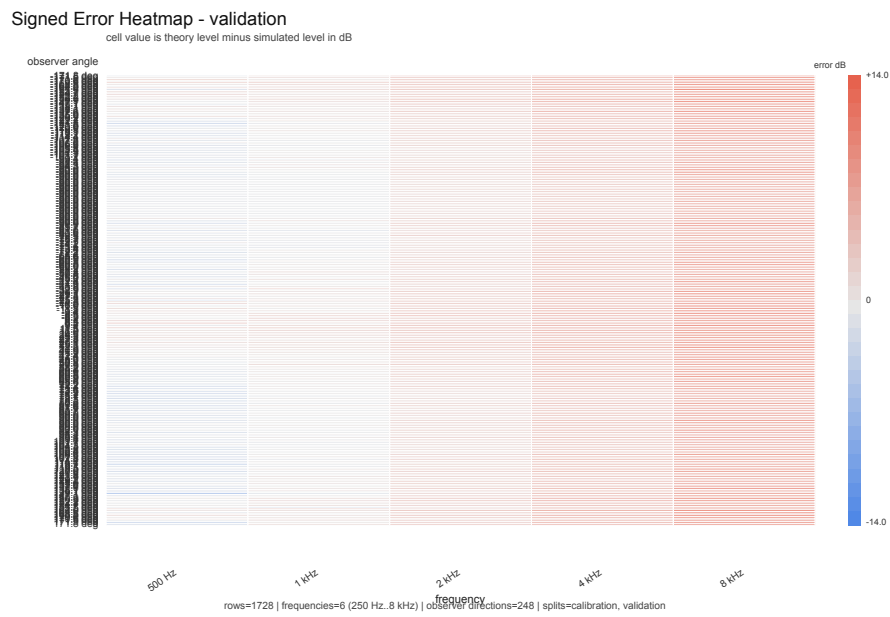


Figure 17: Representative validation error heat map for the same high-resolution case. The residual structure is frequency dependent and directional, reinforcing the interpretation that the remaining mismatch is not a single global amplitude offset.

G Nomenclature

S_{pp}	Far-field pressure power spectral density (PSD)
$\mathbf{C}_q$	Source cross-spectral density matrix
$\mathbf{\Gamma}$	Spatial coherence matrix
$\mathbf{W}$	Integrated radiation operator, $\sum_o w_o \mathbf{a}_o \mathbf{a}_o^H$
$\lambda_r, \mathbf{u}_r$	Eigenvalue and eigenvector of source covariance mode r
$\mathcal{P}_0$	Raw off-diagonal coherence availability index
$\mathcal{P}_R$	Radiation-weighted phase-steerability index
$\mathcal{R}$	Radiating covariance fraction
D_{control}	Numerical dimension of locally controllable directivity subspace
Q_T	Target projection onto the locally controllable directivity subspace
Γ_{mn}	Inter-feather spatial coherence function
Φ_{qq}^{ii}	Local loading-noise autospectrum for feather i
Φ_m	Discretized source autospectrum at source point m
C_Φ	Scalar source strength coefficient
H_i	Acoustic transfer function from feather i to observer
$\mathbf{n}$	Unit observer direction vector
ω	Angular acoustic frequency
k	Acoustic wavenumber (ω/c_0)
c_{ref}	Reference chord used to nondimensionalize geometry and frequency
κ	Acoustic size parameter, $\kappa = kc_{\text{ref}}$
α_i	Local incidence angle for feather i
b_α	Convective delay sensitivity
Π_α	Nondimensional delay scale, $\Pi_\alpha = U_\infty b_\alpha / c_{\text{ref}}$
Θ_{mn}	Nondimensional incidence-delay phase difference, $\omega b_\alpha (\alpha_m - \alpha_n)$
St_0	Fixed Strouhal scale parameter
$\tilde{d}_{mn}$	Coherence-scaled nondimensional separation between source points m and n
U_c	Convective velocity of turbulent eddies
δ^*	Boundary layer displacement thickness
η	Non-dimensional spanwise parametric coordinate ($\eta \in [0, 1]$)
$\mathbf{l}_m$	Unsteady lift dipole vector at source point m
$\mathbf{r}_m$	Spatial coordinate vector at source point m
N	Total number of feathers in the array
N_b	Number of incidence basis modes
M	Total number of discretized source points across all feathers
M_∞	Free-stream Mach number (U_∞/c_0)
$J_i(\eta)$	Spanwise Jacobian (local feather span)
c_0	Speed of sound
ρ_0	Ambient fluid density

H Evaluation of the PSD Matrix

The following pseudocode outlines the implementation used to evaluate the far-field pressure power spectral density across a set of observer directions and frequencies. It illustrates how the double-summation structure and local spanwise numerical integration are aggregated.

Algorithm 1 Evaluation of Positive-Semidefinite Array Far-Field Pressure Spectrum

Require: Observer direction set $\{\mathbf{n}\}$, Frequency set $\{\omega\}$, Discretized source points $m \in \{1 \dots M\}$ with coordinates $\{\mathbf{r}_m\}$, dipole loading directions $\{\mathbf{l}_m\}$, local chords $\{c_m\}$, local quadrature weights $\{w_m\}$, and local incidences $\{\alpha_m\}$

Ensure: Far-field acoustic pressure power spectral density $S_{pp}(\mathbf{n}, \omega)$

```

1: for each frequency  $\omega$  do
2:   Compute source autospectra vector  $\Phi(\omega) \in \mathbb{R}^M$  where  $\Phi_m(\omega)$  is given by Eq. 26
3:   Form source cross-spectral matrix  $\mathbf{C}_q(\omega) \in \mathbb{C}^{M \times M}$ :
4:   for each source point  $m \in \{1 \dots M\}$  do
5:     for each source point  $n \in \{1 \dots M\}$  do
6:       Compute spatial coherence decay:  $\gamma_{mn} \leftarrow \exp(-\tilde{d}_{mn}f/f_{\text{ref}})$ 
7:       Compute incidence-delay phase:  $\theta_{mn} \leftarrow \omega b_\alpha(\alpha_m - \alpha_n)$ 
8:       Evaluate matrix element:  $C_{q,mn}(\omega) \leftarrow \sqrt{\Phi_m(\omega)\Phi_n(\omega)}\gamma_{mn} \exp(i\theta_{mn})$ 
9:     end for
10:  end for
11:  for each observer direction  $\mathbf{n}$  do
12:    Form transfer weight vector  $\mathbf{a} \in \mathbb{C}^M$  where for each source point  $m$ :
13:       $a_m \leftarrow K_H(\mathbf{n}, \omega) c_m(\mathbf{l}_m \cdot \mathbf{n}) w_m \exp(ik\mathbf{n} \cdot \mathbf{r}_m)$ 
14:    Evaluate quadratic form:  $S_{pp}(\mathbf{n}, \omega) \leftarrow \frac{1}{(4\pi R)^2} \mathbf{a}^H \mathbf{C}_q(\omega) \mathbf{a}$ 
15:  end for
16: end for
17: return  $\{S_{pp}(\mathbf{n}, \omega)\}$ 

```
